Inflation Risk and Yield Spread Changes

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Abstract

Inflation risk explains a significant share of the systematic residual variation in yield spread changes beyond standard credit factors and intermediation frictions. Expected inflation, inflation uncertainty, and inflation cyclicality help account for bond-level movements and the residual common component in corporate credit. The effect is stronger for firms closer to default or less able to absorb cost shocks, such as highly leveraged and lower-rated issuers and firms whose margins are squeezed when producer prices rise, and weaker when debt adjusts more quickly through refinancing or floating-rate borrowing. Together, these patterns link residual spread commonality to heterogeneous exposure to inflation risk.

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I. Introduction

Corporate yield spread changes remain empirically challenging to explain. Collin-Dufresne, Goldstein and Martin (2001) (hereafter CDGM) show that fundamental credit-risk variables play a significant role, yet a substantial amount of unexplained variation persists. Much of this unexplained variation is systematic and tied to a common component, suggesting the presence of aggregate factors alongside standard credit-risk variables. Given the predominance of nominal debt in the U.S. corporate sector and the slow adjustment of firms' leverage, changes in the inflation outlook should affect corporate credit risk by changing the real value of promised payments. This channel is largely absent from standard empirical explanations of corporate bond pricing.

In this paper, I investigate whether inflation risk helps explain the large unexplained systematic variation in yield spread changes. I measure inflation risk using three inflation-related variables, primarily constructed from inflation swap rates: expected inflation, inflation uncertainty, and inflation cyclicality. In yield spread change regressions, these proxies account for more than a quarter of the unexplained variation beyond standard credit-risk and intermediation factors and are significantly related to the common residual component.

These variables are motivated by models of nominal debt and default. In Bhamra, Fisher and Kuehn (2011), Gomes, Jermann and Schmid (2016), Gomes and Schmid (2020), and Bhamra, Dorion, Jeanneret and Weber (2022), inflation affects corporate credit risk because liabilities are fixed in nominal terms while leverage adjusts slowly. Expected inflation therefore changes the real value of promised debt payments and, through this channel, default probabilities and credit spreads. Inflation uncertainty captures a different mechanism. In David (2007) and Kang and Pflueger (2015), defaultable debt contains an option-like component, where greater uncertainty about nominal firm value increases the value of the default option and should widen spreads.[1] The credit-risk implications of inflation also depend on

[1] A more indirect channel is suggested by Fischer (2016). In that setting, long-run inflation uncertainty can lower bond values by distorting investment decisions: price uncertainty delays investment or reallocates it toward less productive projects. As in Baldwin and Ruback (1986), higher uncertainty raises the relative

the macroeconomic state. Inflation news that arrives with stronger expected real activity differs from inflation, or disinflation, that arrives in weak real states, because it implies different effects on real debt burdens and repayment capacity, as emphasized by Campbell, Pflueger and Viceira (2020), Fang, Liu and Roussanov (2025), and Bonelli, Palazzo and Yamarthy (2025).[2] These theories motivate the empirical proxies and discipline the interpretation of the cross-sectional evidence. The effect of inflation risk should be stronger for firms whose balance sheets, debt contracts, and operating cash flows make credit risk more exposed to nominal conditions.

I construct the baseline measures from zero-coupon inflation swaps.[3] Since swaps are quoted at multiple maturities, I use them to construct an inflation swap curve and match each bond's duration to the horizon over which inflation risk is most relevant for its promised cash flows. The expected-inflation proxy is the monthly change in the duration-matched swap rate. Inflation uncertainty is measured as the change in the within-month volatility of the same rate. Inflation cyclicality is measured as the change in the rolling correlation between changes in expected inflation and revisions in expected real GDP growth from Consensus Economics.

I document three main empirical results. First, the inflation-risk proxies are significantly related to bond-level yield spread changes in a sample of U.S. corporate bonds from Academic TRACE covering September 2004 to December 2021. Expected inflation is negatively related to spread changes, consistent with the idea that higher inflation lowers the real burden of outstanding nominal debt. Inflation uncertainty is positively related to spread changes, consistent with a default-option channel in which greater uncertainty about nominal firm value widens spreads. Inflation cyclicality is positive but weaker, consistent with a channel whose effect depends on the macroeconomic state. Taken together, the three proxies explain

value of short-lived assets, all else equal, which can increase default risk and widen yield spreads.

[2]Chen (2010) and Bhamra, Dorion, Jeanneret and Weber (2022) also note that recovery rates tend to be lower in low-inflation economic states.

[3]Inflation swaps are among the most liquid over-the-counter inflation-linked derivatives and are widely used to measure inflation compensation across maturities (e.g., Fleming and Sporn (2013); Fleckenstein, Longstaff and Lustig (2016); Diercks, Campbell, Sharpe and Soques (2023)).

between one quarter and one third of the variation left by standard credit-risk variables and recent intermediation controls following Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024) (henceforth FN, HKS, and EHL). The economic magnitudes are also meaningful. A one-standard-deviation increase in expected inflation narrows yield spreads by about 8 basis points, while a one-standard-deviation increase in inflation uncertainty widens them by about 5 basis points.

Second, inflation risk accounts for a significant fraction of the systematic residual variation in spread changes. Following CDGM, I regress monthly yield spread changes on variables motivated by structural credit-risk models. In recent U.S. corporate bond data, these variables explain an important but limited share of monthly spread changes, with a mean adjusted R^2 of 35.5%. The remaining variation is highly systematic, with the first principal component explaining 79.7% of systematic residual variation after aggregating residuals into maturity-leverage cohorts. Projecting this common component onto the inflation-risk proxies shows that inflation risk explains about 21% of the variation in the first principal component of the CDGM residuals. This relation remains economically meaningful after controlling for intermediation, macroeconomic, monetary-policy, and growth-expectation variables.

Third, I document large heterogeneous effects across firms, consistent with inflation risk operating through balance sheets, debt contracts, and operating cash flows. If expected inflation affects credit risk by changing the real value of nominal liabilities, the effect should be stronger when default risk is more sensitive to the debt burden. Consistent with this mechanism, spreads are more sensitive to inflation risk for firms with high leverage, weak profitability, low profit margins, and lower credit quality. These are firms for which a change in the real value of promised debt payments is more likely to affect distance to default. Debt-contract characteristics point in the same direction. Firms with more floating-rate debt or more frequent refinancing needs are less exposed to inflation risk, since nominal debt adjusts more quickly and weakens the real debt-burden channel.[4] I also decompose the residual

[4]The floating-rate debt and refinancing variables proxy for how quickly nominal debt adjusts. The mechanism does not require direct indexation to inflation; firms with faster liability repricing are simply less

variation explained by the inflation-risk proxies to identify which dimension of inflation risk carries the explanatory power across these groups. Expected inflation explains the largest share of residual variation where nominal liabilities are large or slow to adjust, including the leverage, credit-rating, recent debt-growth, and refinancing sorts.

The cross-sectional evidence also points to a cash-flow channel alongside the liability channel. Inflation affects credit risk not only through the real value of debt, but also through input costs, output prices, and operating cash flows. I proxy pricing power with markup, since firms with higher markups are better able to absorb or pass through cost shocks. I capture exposure to cost pressures with input-cost exposure, which reflects how sensitive an industry's cash flows are to inflation in production inputs.[5] Firms with low markups are more sensitive to inflation risk, while firms with greater input-cost exposure respond more strongly. In the variance decomposition, these operating-exposure measures are especially related to inflation uncertainty, suggesting that uncertainty about the inflation outlook matters more when cash flows are more exposed to cost pressures. Inflation cyclicality plays a smaller role overall, but becomes more visible when the expected-inflation channel is muted.

The evidence is robust across measurement choices, specifications, and inflation regimes. Results are similar when using survey-anchored expectations, constructed by fitting a daily Nelson-Siegel-Svensson curve following Reis (2020) to swaps, TIPS breakevens, and survey data. The main signs also hold across fixed-maturity swaps, TIPS breakevens, CPI-based proxies, realized inflation controls, forecast disagreement, firm-level equity volatility, and additional aggregate controls. The results are robust to alternative residual groupings, panel specifications, clustering schemes, and an extended sample through 2024 using Enhanced TRACE. The time-series evidence points in the same direction: expected inflation matters more when inflation risk is salient, inflation uncertainty becomes more important when the

exposed to changes in the real value of fixed nominal debt.

[5] Input-cost exposure is constructed at the industry level using Producer Price Index data and input-output linkages. I first measure variation in an industry's own output prices and then adjust it for the price flexibility of supplier industries, weighted by their importance in production. Higher values indicate that cash flows are more closely tied to sectoral price movements and input-cost pressures.

outlook is harder to forecast, and inflation cyclicality varies across regimes because its effect depends on whether inflation news arrives with stronger or weaker expected real activity.

Taken together, the evidence shows that inflation risk explains part of the variation in yield spread changes left by standard credit-risk and intermediation variables. This explanatory power is not uniform across firms. Expected inflation matters most when nominal liabilities are large or slow to adjust, while inflation uncertainty matters more when debt terms, maturity, or operating cash flows make promised payments harder to value. These results link residual spread commonality to nominal debt, debt-contract flexibility, and firms' exposure to inflationary cost pressures.

This paper contributes to three related strands of the literature. First, it contributes to the literature on inflation and corporate credit. A large asset-pricing literature studies inflation as an aggregate risk factor in equity and bond markets, including Fama (1981), Chen, Roll and Ross (1986), Weber (2014), Eraker, Shaliastovich and Wang (2015), Fleckenstein, Longstaff and Lustig (2017), and Boons, Duarte, de Roon and Szymanowska (2020). In corporate credit, existing work shows that inflation affects default risk and bond prices through nominal liabilities, inflation uncertainty, and the relation between inflation and real activity, as in David (2007), Kang and Pflueger (2015), Bhamra, Fisher and Kuehn (2011), Gomes, Jermann and Schmid (2016), Gomes and Schmid (2020), and Bhamra, Dorion, Jeanneret and Weber (2022). More recent work studies related channels in corporate bond returns, including Ceballos (2021) and Lu, Nozawa and Song (2025). I add to this literature by showing that inflation risk helps explain variation in individual corporate bond yield spread changes, including the residual variation left by standard credit-risk variables.

Second, the paper contributes to the literature on the common component in corporate bond spread changes. CDGM show that standard credit-risk variables leave a large common residual component in yield spread changes. Other work attributes this residual commonality to liquidity, market structure, and intermediary constraints, including Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). I show that inflation risk explains part of this common residual component, linking residual spread commonality to aggregate nominal risk as well as to market frictions and intermediation.

Third, the paper contributes to the literature on the transmission of inflation risk across firms. The nominal-debt literature shows that inflation affects default risk by changing the real value of promised debt payments, as in Bhamra, Fisher and Kuehn (2011), Gomes, Jermann and Schmid (2016), Gomes and Schmid (2020), and Bhamra, Dorion, Jeanneret and Weber (2022), while theories of debt maturity and refinancing emphasize how contract structure shapes firms' sensitivity to shocks, including Diamond (1991), Goldstein, Ju and Leland (2001), and Friewald, Nagler and Wagner (2022). I bring these mechanisms to bond-level yield spread changes and show that the effects are stronger for firms whose balance sheets, debt contracts, and operating cash flows make credit risk more exposed to nominal conditions. The variance decomposition further shows that expected inflation explains more residual variation along balance-sheet and debt-contract margins, while inflation uncertainty is more important where maturity, liability repricing, or operating exposure make promised payments harder to value.

The remainder of the paper is organized as follows. Section II outlines the data sources. Section III introduces the inflation-risk proxies, presents the main results, and reports the survey-anchored specifications. Section IV discusses the heterogeneity results. Section V provides additional evidence and robustness tests. Section VI concludes.

II. Data

I rely on several data sources to analyze the impact of inflation risk on yield spread changes. The sample of industrial corporate bond transactions comes from the Academic Trade Reporting and Compliance Engine (TRACE), maintained by the Financial Industry Regulatory Authority (FINRA). The main sample covers September 2004 to December 2021. In the robustness analysis, I extend the sample through December 2024 using the Enhanced TRACE database.[6] I follow the cleaning steps from Dick-Nielsen and Poulsen (2019), thus cleaning same-day corrections and cancellations, removing reversals, as well as double counting of agency trades. Then, I apply a median filter and a reversal filter to eliminate further potential data errors following Edwards, Harris and Piwowar (2007). The median filter identifies potential outliers in reported prices within a specific time period, while the reversal filter captures unusual price movements.[7] I merge corporate bond pricing data with the Mergent Fixed Income Securities Database (FISD) to obtain bond characteristics, such as offering amount, offering date, maturity, coupon rate, bond rating, bond option features, and issuer information, and with CRSP/Compustat to obtain firm characteristics.[8]

Following the literature on corporate bonds, I restrict the sample to corporate debentures and exclude bonds with variable coupons, convertibility, putability, asset-backed status, exchangeability, private placements, perpetual terms, preferred securities, secured lease obligations, missing ratings, or foreign-currency denomination. I also remove bonds issued by financial firms (SIC 6000–6999) and utility firms (SIC 4900–4999), and bonds with issue sizes under \$10 million or a time to maturity of more than 30 years or less than one month.[9]

The main variable in the empirical analysis is the monthly change in the corporate bond yield spread. Using TRACE intraday data, I first eliminate transactions with when-issued, lock-in, special trades, or primary trades flags. I then calculate the daily clean price as the volume-weighted average of intraday prices to reduce the effect of bid-ask bounce and individual trade noise, following Bessembinder, Kahle, Maxwell and Xu (2008). For each

[6] The main sample ends in December 2021 because the Enhanced TRACE database lacks dealer identifiers and therefore does not permit the construction of the full set of OTC market proxies used in the baseline specifications. In the extended sample, I use only the subset of OTC variables that can be constructed without dealer identifiers.

[7] The median filter eliminates any transaction where the price deviates by more than 10% from the daily median or from a nine-trading-day median centered at the trading day. The reversal filter eliminates any transaction with an absolute price change that deviates from the lead, lag, and average lead/lag price change by at least 10%.

[8] See the Internet Appendix for the construction of the TRACE/CRSP linking table. The matching uses issuer CUSIPs, parent CUSIPs, trading symbols, and firm names, and accounts for mergers, delistings, and ticker changes.

[9] The results remain robust when all bonds, irrespective of industry or bond type, are included, as reported in the Internet Appendix.

month, I use the observation closest to the last trading day of the month within a five-trading-day window.[10] I compute the end-of-month corporate bond yield from the volume-weighted price and define the yield spread as the difference between this yield and the yield of a risk-free bond with the same cash-flow structure as the corporate bond. The risk-free benchmark is derived from the U.S. Treasury yield curve estimates published by the Federal Reserve Board.[11] I compute monthly changes and returns of all variables. To avoid asynchronicity issues, I match the dates of variables available at the daily frequency, such as the VIX and Treasury rates, to the dates on which end-of-month bond prices are measured. I retain only bonds with at least 25 observations of monthly yield spread changes.

Following CDGM, I construct market and firm-specific variables that, according to structural models, determine yield spread changes. These include the Standard & Poor's 500 index return (RM_t) from CRSP, changes in the VIX index (ΔVIX_t) from the Chicago Board Options Exchange, and changes in Treasury rates and the term-structure slope (ΔRF_t, ΔRF_t^2, and $\Delta Slope_t$) from the off-the-run yield curves of Gürkaynak, Sack and Wright (2007). As a systematic proxy for the probability or magnitude of a downward jump in firm value ($\Delta Jump_t$), I construct a measure based on at- and out-of-the-money put options and at- and in-the-money call options with maturities of less than one year, traded on the SPX index. The option data come from OptionMetrics and the Options Price Reporting Authority (OPRA). For the exact procedure used to estimate the jump component, I refer to CDGM.

I use market leverage as a proxy for firm creditworthiness. Following Friewald and Nagler (2019), market leverage ($\Delta Lev_{i,t}$) is book debt divided by the sum of book debt and the market value of equity. Book debt is Compustat long-term debt (DLTT) plus debt in current liabilities (DLC). I lag debt-related accounting information by six months to avoid using data before it is plausibly available to investors. Market equity is common shares outstanding times the CRSP share price.[12]

[10] Using a five-day window increases coverage while keeping the price close to month-end. The results are robust to using only bonds with an end-of-month price, as reported in the Internet Appendix.

[11] Yield spreads are winsorized at the 0.5% level.

[12] Using a six-month lag avoids using accounting information before it is plausibly available to investors.

The inflation-related variables are constructed from inflation-linked market prices and survey expectations. I obtain daily bid and ask quotes for U.S. zero-coupon inflation swaps from Bloomberg for maturities between one and thirty years starting in July 2004.[13] Inflation swaps are forward contracts in which the inflation buyer pays a fixed nominal rate and receives a payment linked to realized inflation. They provide market-based inflation compensation across maturities and are the main source for the baseline inflation measures.

I also use zero-coupon TIPS yields and breakeven inflation rates from Gürkaynak, Sack and Wright (2010), which are derived from TIPS coupon bond yields. Since both inflation swaps and TIPS are indexed to the non-seasonally adjusted CPI-U, I seasonally adjust both curves following Fleckenstein, Longstaff and Lustig (2014).[14] For each corporate bond, I match the bond's duration to the corresponding point on the inflation curve, using spline interpolation when the required maturity is not directly quoted. Section III defines the three baseline inflation-risk proxies constructed from this duration-matched curve.

I also construct survey-anchored inflation-expectation measures. I use the Michigan Survey of Consumers, the Survey of Professional Forecasters, and Consensus Economics inflation forecasts. Following Dovern, Fritsche and Slacalek (2012), I convert current-year and next-year Consensus Economics forecasts into fixed 12-month-ahead expectations by weighting the two calendar-year forecasts according to the share of the forecast horizon falling in each year.[15] I combine inflation swaps, TIPS breakevens, and survey expectations in a Nelson-Siegel-Svensson curve fitted daily to the inflation term structure. Survey observations are aligned to their publication dates and carried forward between releases using exponential time-decay weights.[16] Evaluating the fitted curve at each bond's duration yields a bond-

[13] I use the liquid benchmark maturities from one to ten years, and the 12-, 15-, 20-, and 30-year maturities. Bloomberg does not retain inflation swap quotes before July 2004. I use mid quotes, computed as the average of bid and ask quotes.

[14] Seasonal adjustment starts at the shortest available maturity, one year for zero-coupon inflation swaps and two years for TIPS breakevens. This matters because duration matching may evaluate the inflation curve at fractional maturities, where unadjusted rates can mechanically reflect seasonal CPI patterns. The full procedure is described in the Internet Appendix.

[15] This avoids discontinuities that arise from using calendar-year forecasts directly.

[16] In the baseline implementation, the time-decay half-life is 30 days for the Survey of Professional Forecasters and 15 days for the Michigan Survey of Consumers and Consensus Economics.

specific duration-matched expected-inflation series. This provides a complementary measure of inflation expectations that does not rely on any single market or survey series. For the cyclicality measure, I also use Consensus Economics forecasts of real GDP growth.

Lastly, Producer Price Index (PPI) data and input-output tables, used to estimate industry-level input-cost exposure, come from the U.S. Bureau of Labor Statistics, while data on realized inflation, unemployment, real consumption, and income come from the Federal Reserve Bank of St. Louis. I construct intermediation proxies using intraday Academic TRACE data following FN and EHL. To construct the distress factor of HKS, I use the intermediary capital ratio from Zhiguo He's website and the noise measure from Jun Pan's website. For the exact procedures, I refer to Friewald and Nagler (2019), He, Khorrami and Song (2022), and EHL.

Table I Panel A reports the summary statistics of the sample of corporate bonds. The sample consists of 449788 observations of monthly yield spread changes for 6,826 bonds issued by 936 firms. The average yield spread is 2.32%, with a standard deviation of 3.05%. The average offering size is $742 million, and the average time to maturity is approximately nine years. Around 21% of the observations are high-yield bonds.

Table I about here

III. Inflation Risk and Yield Spread Changes

This section examines whether inflation risk helps explain the residual component in corporate yield spread changes. The starting point is the benchmark evidence in CDGM. Standard credit-risk variables explain an important part of spread movements, but leave a large residual component that is highly systematic across bonds. I use this residual component as the object of interest and ask whether it is related to expected inflation, inflation uncertainty, and inflation cyclicality.

I first introduce the inflation-risk proxies and the channels they are meant to capture. I then estimate their relation to yield spread changes after controlling for the standard credit-risk variables of CDGM and the intermediation factors of FN, HKS, and EHL. Next, I examine whether the results are sensitive to the measurement of inflation expectations using survey-anchored term-structure estimates. Finally, I assess the robustness of the findings under alternative estimation and inference procedures.

A. *Inflation-Risk Proxies*

Inflation expectations should be measured at the horizon that matters for each bond. A given inflation-horizon shock should not affect short- and long-duration bonds equally. I therefore construct the main baseline expected-inflation and uncertainty proxies from duration-matched inflation swap rates. Inflation swaps provide forward-looking, market-based inflation compensation across maturities, which allows me to match each bond's duration to the corresponding point on the inflation curve. I complement these bond-level measures with a time-series proxy for inflation cyclicality, constructed from the relation between long-term expected inflation and expected real GDP growth. Together, these variables capture three inflation-related dimensions: expected inflation, inflation uncertainty, and the macroeconomic state in which inflation news arrives.

Inflation swaps are well suited for the baseline analysis because they are observed at high frequency and provide inflation rates across maturities. I therefore use them as the main source of forward-looking inflation variation. Since traded inflation-linked instruments may also contain risk-premium and liquidity components, Section III.C reports a complementary analysis using survey-anchored Nelson-Siegel-Svensson fitted expectations. This exercise checks whether the main results are specific to raw swap rates while preserving the maturity-specific information that makes market prices useful.

Expected inflation. The first proxy captures innovations in expected inflation. The channel follows from the nominal nature of corporate debt. Corporate bonds promise fixed nominal payments, while firms adjust leverage only gradually. Changes in expected inflation therefore affect the real value of promised liabilities and, through this channel, default risk. Higher expected inflation reduces the real burden of outstanding debt, lowers real leverage, and should compress yield spreads. This prediction follows from models with nominal debt and sticky leverage, including Bhamra, Fisher and Kuehn (2011), Gomes, Jermann and Schmid (2016), Gomes and Schmid (2020), and Bhamra, Dorion, Jeanneret and Weber (2022). Conversely, declines in expected inflation raise the real burden of debt and can strengthen debt-overhang forces. In Gomes, Jermann and Schmid (2016), this mechanism distorts investment and production decisions and raises credit spreads.

Expected inflation can also coincide with changes in expected cash flows. Over much of the sample, expected inflation is positively related to expected real activity, so increases in expected inflation often occur together with stronger expected fundamentals (David and Veronesi (2013); Campbell, Pflueger and Viceira (2020); Bonelli, Palazzo and Yamarthy (2025)).[17]

I measure expected-inflation innovations with the monthly change in the duration-matched inflation swap rate, denoted by $\Delta E[\mu^S]_{i,t}$. For each bond, I evaluate the zero-coupon inflation swap curve at the bond's duration and use the resulting rate as the relevant inflation horizon. This procedure matches short-duration bonds to short-horizon inflation compensation and long-duration bonds to longer-horizon measures. The expected coefficient on $(\Delta E[\mu^S]_{i,t})$ is negative.[18]

Inflation uncertainty. The second proxy captures uncertainty about the inflation outlook. Its effect differs from the expected-inflation channel. While higher expected inflation lowers the real value of nominal debt, greater uncertainty about future inflation increases

[17] Bonelli, Palazzo and Yamarthy (2025) show that the sign of inflation sensitivity depends on the relation between inflation and expected growth. In the full sample, expected inflation is, on average, positively related to expected growth, so the unconditional expected-inflation coefficient is predicted to be negative.

[18] Because measured leverage enters directly in the benchmark regressions, the coefficient on expected inflation captures variation in spreads that is not summarized by contemporaneous leverage alone. Expected-inflation innovations can affect the real value of outstanding liabilities, expected nominal cash flows, refinancing conditions, and default risk before these effects are fully reflected in measured leverage. This distinction is important because the empirical tests ask whether the inflation-risk proxies add explanatory power beyond the standard CDGM controls, including measured leverage.

the dispersion of nominal firm value. In David (2007) and Kang and Pflueger (2015), defaultable debt can be viewed as a risk-free bond minus a put option on firm value. Higher uncertainty raises the value of this default option and widens spreads. A related channel operates through investment. In Fischer (2016), long-run inflation uncertainty lowers bond values by delaying or distorting investment decisions; as in Baldwin and Ruback (1986), higher uncertainty can make short-lived assets relatively more valuable, reducing the value of longer-lived claims and increasing yield spreads.

I measure this channel with the change in the monthly standard deviation of the duration-matched inflation swap rate, denoted by $\Delta\sigma^S_{i,t}$. For each bond-month, I compute the standard deviation of the daily matched swap rate within the month and then take the monthly change. The measure captures instability in the market-implied inflation path rather than shifts in the level of expected inflation. The expected coefficient on $\Delta\sigma^S_{i,t}$ is positive. [19]

Inflation cyclicality. The third proxy captures the macroeconomic state in which inflation news arrives. Inflation has different credit-risk implications depending on whether it is associated with stronger or weaker expected real activity. Inflation news that arrives with stronger expected real activity differs from inflation, or disinflation, that arrives in weak real states because it implies different effects on real debt burdens and repayment capacity. This state dependence is emphasized by Kang and Pflueger (2015), Bhamra, Dorion, Jeanneret and Weber (2022), Campbell, Pflueger and Viceira (2020), Fang, Liu and Roussanov (2025), and Bonelli, Palazzo and Yamarthy (2025).

I capture this channel with an expectations-based cyclicality proxy, denoted by $\Delta\rho^S_t$. The level (ρ^S_t) is the rolling 24-month correlation between changes in long-horizon expected inflation and revisions in expected real GDP growth.[20] Expected growth revisions are measured

[19]This proxy is distinct from forecast disagreement. Disagreement measures dispersion in beliefs across forecasters, while $\Delta\sigma^S_{i,t}$ measures time-series variation in the market-implied inflation outlook. I use disagreement-based measures as part of the robustness analysis.

[20]For the cyclicality proxy, expected inflation is measured using the 10-year inflation swap rate. I use the 10-year zero-coupon inflation swap rate when available and the 10-year TIPS breakeven rate otherwise. The results are similar when expected inflation is instead measured using the Nelson-Siegel-Svensson fitted

using the Consensus Economics fixed-horizon 12-month-ahead GDP forecast.

The proxy ρ_t^S captures the hedge value of inflation for nominal corporate debt. A higher correlation between expected inflation and expected growth means that weak growth states are more likely to coincide with lower inflation and, therefore, with a higher real value of nominal debt. An increase in ρ_t^S, and hence in $\Delta\rho_t^S$, should therefore be associated with wider spreads. Unlike expected inflation and inflation uncertainty, this channel is inherently state-dependent. A pooled coefficient may average across regimes in which inflation has different credit-risk implications, so I interpret the cyclicality estimates together with the time-series evidence in Section V.D.

Table I Panel B reports the unconditional correlations between yield spread changes and the three inflation-risk proxies, as well as the correlations among the proxies.

B. *Baseline Results*

I begin by documenting the residual variation in recent U.S. corporate bond data. Following CDGM, I estimate a separate time-series regression for each bond,

$$\Delta YS_{i,t} = \alpha_i + \boldsymbol{\beta}_i^{\boldsymbol{T}} \boldsymbol{\Delta S}_{i,t} + \varepsilon_{i,t},$$

where $\Delta YS_{i,t}$ is the monthly yield spread change and $\boldsymbol{\Delta S}_{i,t}$ contains the standard structural credit-risk controls. These controls include changes in leverage, Treasury rates, squared changes in Treasury rates, changes in the slope of the term structure, changes in the VIX, the S&P 500 return, and the jump-risk proxy.

Column (1) of Table II reports the benchmark results.[21] The mean adjusted R^2 is 35.5%. Thus, standard credit-risk variables explain an important part of monthly spread movements, but close to two-thirds of the variation remains unexplained. This magnitude is in line with

Table II about here

inflation curve described in Section III.C.

[21] Detailed coefficient estimates for the CDGM replication and the corresponding cohort-level PCA loadings are reported in the Internet Appendix.

the evidence in CDGM and with recent work.

The residual variation is also highly systematic. To measure this commonality, I assign bonds to 18 maturity-leverage cohorts. The maturity groups are below five years, five to eight years, and above eight years; the leverage groups are below 15%, 15%–25%, 25%–35%, 35%–45%, 45%–55%, and above 55%. For each cohort and month, I compute the average residual $\boldsymbol{\varepsilon}_{g,t}$ and extract principal components from the covariance matrix of the cohort-level residuals.[22]

The principal component analysis confirms that this residual variation is highly concentrated in one common component. The total unexplained variance, measured by the trace of the covariance matrix, is about 112 basis points. The first principal component accounts for 79.7% of systematic residual variation, while the second accounts for only 5.4%. The residuals are therefore not simply idiosyncratic bond-level noise. They contain a common component that remains after standard credit-risk variables have been removed. I use this common component to test whether inflation risk accounts for part of the systematic variation left unexplained by the benchmark credit-risk specification.

I add the inflation-risk proxies to the benchmark specification:

$$\Delta YS_{i,t} = \alpha_i + \boldsymbol{\beta}_i^T \boldsymbol{\Delta S}_{i,t} + \boldsymbol{\theta}_i^T \boldsymbol{\Delta I}_{i,t} + \boldsymbol{\Gamma}_i^T \boldsymbol{\Delta C}_{i,t} + \nu_{i,t},$$

where $\boldsymbol{\Delta I}_{i,t}$ contains $\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, and $\Delta\rho^S_t$. The vector $\boldsymbol{\Delta C}_{i,t}$ contains the OTC market friction controls from FN, the intermediary distress factors from HKS, and interdealer price dispersion from EHL.

Panel A of Table II reports average bond-level coefficients, t-statistics, and mean and median R^2 values. Following CDGM, inference is based on the cross-sectional dispersion of the bond-level estimates. I compute the t-statistic as the average coefficient divided by the standard deviation of the bond-level coefficients, scaled by the square root of the number of

[22]The PCA is performed on cohort-month average residuals rather than on individual bond residuals, following CDGM. This reduces the influence of idiosyncratic bond-level noise. Section V shows that the results are similar under alternative cohort definitions.

bonds.[23] Columns (2) to (4) add the inflation-risk proxies one at a time, column (5) includes them jointly, and columns (6) to (8) add the intermediation controls.

The signs are stable across specifications. Increases in expected inflation are associated with narrower spreads, and increases in inflation uncertainty with wider spreads. The cyclicality proxy also enters positively, but the estimate is weaker than those for expected inflation and uncertainty. This is consistent with the proxy capturing regime-dependent variation that a single full-sample coefficient cannot summarize well.

I use three residual measures in the analysis. The first is the bond-level residual from the spread-change regressions. The second is the cohort-level residual, which averages these residuals within the 18 maturity-leverage cohorts. This is the object used to compute the fraction of residual variation explained. The third is the common residual component, measured as the first principal component of the cohort-level residuals. In each case, explanatory power is measured relative to the corresponding specification without the inflation-risk proxies. When intermediation controls are included, the comparison is therefore to the residual variance left after adding those controls, but before adding expected inflation, inflation uncertainty, and inflation cyclicality.

The inflation-risk proxies account for a sizeable share of the residual variation. Following He, Khorrami and Song (2022), I measure explanatory power using the fraction of variation explained, defined as the reduction in residual variance after adding the inflation-risk proxies. For each cohort g, let $\varepsilon_{g,t}$ denote the residual from the specification without inflation-risk proxies and $\nu_{g,t}$ the residual from the corresponding specification that includes them. The fraction of variation explained is

$$\text{FVE} = 1 - \frac{\sum_{g=1}^{18} \sigma^2_{\nu_g}}{\sum_{g=1}^{18} \sigma^2_{\varepsilon_g}}, \tag{1}$$

[23] This procedure is standard in this literature, including CDGM, FN, HKS, and EHL. Since the sample contains 6826 bonds, the resulting t-statistics can be large in absolute value. Section III.D verifies the main results using panel fixed-effects regressions with alternative clustering schemes.

where $\sigma^2_{\nu_g}$ and $\sigma^2_{\varepsilon_g}$ are the time-series variances of the residuals for cohort g in the specifications with and without inflation-risk proxies, respectively.

The three inflation-risk proxies jointly explain 36.6% of residual variation before adding intermediation controls. Their explanatory power remains economically large after controlling for OTC frictions, intermediary distress, and interdealer price dispersion. In the fully controlled specification in column (8), they explain 27.0% of the remaining residual variation. Thus, inflation risk explains a separate component of residual spread variation beyond the intermediation factors emphasized in recent work.

Panel C of Table II turns to the common component of the residuals. I regress the first principal component of the CDGM residuals on the inflation-risk proxies.[24] The inflation-risk proxies are jointly significant and explain 21.2% of the variation in PC1 before adding intermediation controls, and 5.8% in the fully controlled specification. Inflation risk therefore helps explain not only bond-level residual variation but also the common residual component.

The economic magnitudes are also substantial. I use the full specification in column (8) and compute the implied yield spread change from a one-standard-deviation change in each inflation proxy. A one-standard-deviation increase in expected inflation, $\Delta E[\mu^S]_{i,t}$, narrows spreads by 8.2 basis points. A one-standard-deviation increase in inflation uncertainty, $\Delta\sigma^S_{i,t}$, widens spreads by about 5.2 basis points, while the effect of inflation cyclicality, $\Delta\rho^S_t$, is smaller, at about 0.6 basis points. These effects are sizeable. The impact of expected inflation is close to that of a one-standard-deviation change in the 10-year Treasury rate, which is around 9 basis points. It is also large relative to average spreads: a one-standard-deviation increase in the swap rate reduces the mean yield spread by 3.5% and the median yield spread by 5.8%.

[24] Because PC1 is a monthly time-series factor, I use monthly cross-sectional averages of expected inflation and inflation uncertainty. The results are similar when fixed-maturity inflation swap rates are used instead.

C. Survey-Anchored Inflation Expectations

The baseline analysis uses duration-matched inflation swap rates because swaps provide high-frequency, forward-looking inflation compensation across maturities. This term-structure information is useful for matching inflation exposure to each bond's duration. At the same time, swap rates are traded prices and may embed risk premia and liquidity components in addition to expected inflation. I therefore construct a complementary measure that preserves the maturity information in market prices while anchoring the curve with survey expectations.

Following the approach in Reis (2020), I combine market-based inflation compensation with survey expectations in a daily fitted inflation curve. Specifically, I fit a Nelson-Siegel-Svensson curve, following Nelson and Siegel (1987) and Svensson (1994), to inflation swap rates, TIPS breakevens, and inflation forecasts from the Michigan Survey of Consumers, the Survey of Professional Forecasters, and Consensus Economics. Because the surveys are released on different dates and at different frequencies, I align each observation to its publication date and carry it forward between releases using exponential time-decay weights. More recent survey readings therefore receive more weight in the daily curve.[25]

Evaluating the fitted curve at each bond's duration gives a daily, bond-specific measure of expected inflation. From this fitted series, I construct the same inflation-risk proxies used in the swap-based analysis. The first, $\Delta E[\mu^E]_{i,t}$, is the monthly change in fitted expected inflation. The second, $\Delta\sigma^E_{i,t}$, is the change in the rolling 21-day volatility of the fitted series. The third, $\Delta\rho^E_t$, is the change in the rolling 24-month correlation between fitted expected-inflation changes and revisions in the Consensus Economics GDP forecast.

Table III reports the results. The table follows the same structure as Table II, but replaces the swap-based proxies with the survey-anchored measures. The signs are unchanged. Expected inflation remains negatively related to yield spread changes, while inflation uncer-

Table III about here

[25] In the baseline implementation, the half-life is 30 days for the Survey of Professional Forecasters and 15 days for the Michigan Survey of Consumers and Consensus Economics.

tainty remains positively related. The cyclicality proxy also enters positively.

The explanatory power is close to the swap-based benchmark. Jointly, the survey-anchored proxies explain 24.6% of residual variation, compared with 27.0% in the baseline swap specification. The point estimates are somewhat smaller, consistent with survey-anchored expectations being smoother and containing less high-frequency variation than market prices. The relation between inflation risk and yield spread changes therefore survives when the market-based term structure is anchored with survey information.

D. Estimation Approach and Inference

In the baseline estimates in Tables II and III, I estimate a separate time-series regression for each bond and then average the coefficients across bonds. This approach allows each bond to have its own sensitivity to the credit-risk controls and inflation-risk proxies, which is useful in a market with substantial heterogeneity across issuers and bond characteristics. Statistical significance is computed from the cross-sectional dispersion of the bond-level estimates, following CDGM. As EHL note, however, the bond-level time-series betas can be noisy, which may affect the standard errors of the average coefficients. Since the sample also contains several thousand bonds, the resulting t-statistics can be large in absolute value.

Table IV examines whether the main conclusions depend on this estimation approach. Column (1) repeats the mean-group estimates from column (8) of Table II. Columns (2) to (4) pool all bond-month observations in panel fixed-effects regressions with bond fixed effects. These specifications impose common slope coefficients across bonds and differ only in the clustering of standard errors. Column (2) double-clusters by bond and month, column (3) clusters by month, and column (4) clusters by bond.[26]

Table IV about here

The point estimates are stable across the two approaches. The expected-inflation coefficient is -0.328 under the mean-group estimator and -0.340 under panel fixed effects. The corresponding coefficients on inflation uncertainty are 0.716 and 0.587. Expected infla-

[26] The panel fixed-effects estimator is less flexible than the mean-group estimator because it imposes homogeneous slopes across bonds. I therefore use it as an inference check rather than as the main specification.

tion and inflation uncertainty remain statistically significant under double clustering. The cyclicality proxy remains positive, although its significance depends more on the inference procedure. This is consistent with cyclicality being state-dependent and less well summarized by a single pooled coefficient.

Overall, the results in this section show that inflation risk explains a sizeable fraction of the systematic residual variation left by standard credit-risk and intermediation controls.

IV. Heterogeneity of Inflation Risk

The baseline results show that inflation risk explains a sizeable share of residual yield spread variation. This section studies how this relation varies across firms and which dimensions of inflation risk account for the explained variation. I distinguish between sensitivity and importance. Sensitivity refers to how strongly spreads respond to expected inflation, inflation uncertainty, and inflation cyclicality. Importance refers to how much each proxy contributes to residual spread variation. The two need not coincide, since a proxy with a large coefficient may contribute little if it varies little in the data.

The cross-sectional tests focus on three features that shape the exposure of credit risk to inflation. The first is balance-sheet strength. When repayment capacity is weak, changes in the real value of nominal debt should have larger effects on default risk, implying stronger spread responses for firms with lower profitability, lower margins, smaller size, or worse ratings. The second is operating exposure. Inflation shocks should be more relevant for firms whose cash flows are more vulnerable to cost pressures or whose pricing power is limited. The third is debt structure. Inflation risk should matter more when nominal liabilities are larger, as for highly levered firms or firms with recent debt growth, and less when debt contracts reset more quickly, as with floating-rate debt, frequent refinancing, or shorter maturity.

This classification separates firms according to whether inflation primarily affects credit risk through the real burden of debt, through operating cash flows, or through the extent to which debt contracts insulate firms from persistent changes in nominal conditions.

These tests follow the logic of models with nominal debt and sticky leverage (Bhamra, Fisher and Kuehn (2011); Gomes, Jermann and Schmid (2016); Gomes and Schmid (2020); Bhamra, Dorion, Jeanneret and Weber (2022)), while extending the evidence to credit quality, pricing power, input-cost exposure, and debt-contract flexibility.

For each industrial bond i with at least 25 monthly observations, I estimate the bond-level time-series regression from Section III,

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

and retain the coefficients on expected inflation, inflation uncertainty, and inflation cyclicality. I then sort bonds along each heterogeneity dimension and compute mean and median coefficients within each group. Bonds are assigned using the average value of the relevant characteristic over the bond's sample life.[27] Figures 1–3 report the resulting patterns for fundamentals and credit quality, operating exposure, and debt structure.

I complement the coefficient estimates with a variance decomposition based on the fraction of variance explained (FVE) measure. For each heterogeneity dimension, I compute how much of the residual variation is explained by each inflation-risk proxy, using a common denominator across groups so that contributions are directly comparable within the dimension.

The decomposition adds the inflation-risk proxies one at a time to the control specification. Group-level contributions are measured as reductions in the variance of group-level residuals, expressed as shares of the summed residual variance across groups within the same dimension.[28] Figure 4 reports the decomposition for all twelve dimensions.

Figures 1, 2, 3, and 4 about here

[27] In unreported tests, the heterogeneity patterns are similar in panel regressions that interact the inflation-risk proxies with the corresponding firm characteristics.

[28] I form portfolios of control-model residuals by group and month, compute the variance of each group's residual series, and express the reduction in this variance from adding a given proxy as a share of the summed across-group residual variance. Because the denominator is common within each dimension, group-level contributions sum to the dimension-level contribution. The single-proxy contributions do not necessarily sum to the joint contribution because the proxies are correlated. In Figure 4, the stacked bars report the

The first set of results supports the liability channel. Figure 1 shows that expected inflation has a stronger effect for firms with weaker fundamentals and lower credit quality. The coefficient is most negative for firms with low return on assets, low profit margins, and lower ratings. Firm size points in the same direction, although the pattern is more moderate. These are firms for which changes in the real value of promised debt payments are more likely to affect distance to default. Inflation uncertainty enters with the opposite sign and follows a similar ordering, as spreads of weaker and lower-rated firms widen more when inflation uncertainty rises. Inflation cyclicality is smaller and less systematically related to these characteristics.[29]

Figure 2 shows that the heterogeneity is not limited to fundamentals and credit quality. The operating sorts help distinguish a purely balance-sheet interpretation from a cash-flow channel in which inflation affects input costs, output prices, and debt-service capacity. The markup split provides the clearest evidence. Low-markup firms have a more negative expected-inflation coefficient than high-markup firms, so their spreads compress more when expected inflation rises. This is consistent with the liability channel, but it also connects naturally to the cash-flow channel, since firms with thinner markups have less room to absorb cost increases through margins. The inflation-uncertainty effect is also larger for low-markup and operationally exposed firms.

Input-cost exposure gives a complementary view from the cost side. Firms with greater input-cost exposure react more strongly to expected inflation, with the largest response concentrated among the most exposed firms.[30] This variable is not simply another proxy for

single-proxy contributions and the diamonds report the joint contribution, so the distance between them captures their shared explanatory power.

[29]In unreported tests, standardized responses preserve the same ordering, suggesting that the heterogeneity is not just a mechanical consequence of higher spread volatility among riskier firms.

[30]Input-cost exposure is constructed at the industry level using Producer Price Index data and input-output linkages. I begin by computing output-price flexibility for each 3-digit NAICS industry as the average absolute change in the industry's Producer Price Index. I then adjust this measure using the output-price flexibility of supplier industries, weighting each supplier industry by the value of intermediate inputs purchased from that industry. Finally, each bond is assigned to a group based on the average input-cost exposure of its issuer's industry over the bond's sample life. Higher values indicate that an industry's cash flows are more closely tied to sectoral price movements and input-cost pressures.

leverage or ratings. It captures whether cash flows are more directly tied to inflationary production costs and sectoral price movements. Operating leverage is less informative, suggesting that the relevant operating margin is not generic fixed-cost intensity, but exposure to inflationary cost pressures and the ability to absorb or pass them through.

I then turn to debt structure, where the evidence more directly targets the real debt-burden channel. The leverage sort in Figure 3 provides the cleanest support. Highly levered firms benefit more from expected inflation because a given increase in the price level erodes a larger stock of real liabilities. Inflation uncertainty follows the same broad pattern, consistent with the default option being more valuable for firms closer to default. Together with the fundamentals and rating sorts, the leverage results indicate that inflation risk matters most when default probabilities are especially sensitive to the debt burden.

Debt-contract flexibility works in the opposite direction. Firms with more distant refinancing needs remain exposed to fixed nominal payment terms for longer, so their spreads respond more strongly to inflation risk. Firms that refinance more frequently can reset nominal terms more quickly, which reduces their exposure. Recent debt growth points in the same direction, with the strongest response among firms in the highest recent debt-growth group. Time to maturity shows a related pattern. The expected-inflation coefficient is most negative for intermediate maturities, while the inflation-uncertainty coefficient rises with maturity and is largest for bonds with more than eight years to maturity. Longer nominal maturity keeps payment terms fixed for longer and increases exposure to uncertainty about the future real value of promised payments.

The final comparison in Figure 3 separates firms with substantial floating-rate debt from firms that predominantly rely on fixed-rate debt. Although the bond sample excludes variable-coupon bonds, the floating-rate measure captures the issuer's broader debt structure. Firms are classified as floating-rate exposed when at least 5% of total debt is floating-rate debt. Their weaker sensitivity indicates that liabilities that reset more quickly dampen the real debt-burden channel. This helps distinguish the inflation channel from a generic aggregate-risk interpretation, since omitted aggregate conditions alone would not predict weaker sensitivity among firms with more flexible debt terms.

The coefficient evidence shows that the inflation-risk proxies load in the predicted directions and that these loadings vary systematically across firms. It does not, however, show which proxy accounts for residual spread variation in each environment. Figure 4 addresses this by decomposing the explained variation into the contribution of each proxy within each group.

The decomposition gives a sharper view of the mechanisms. Expected inflation, which captures the real debt-burden channel, explains the most variation where nominal liabilities are large and slow to adjust. It is the leading contributor in the credit rating, recent debt growth, leverage, refinancing intensity, profitability, and margin sorts, while its contribution is close to zero among the highest-rated issuers. Inflation uncertainty, which captures the default-option channel, explains more variation where the future real value of promised payments is harder to assess or where operating cash flows are more exposed. It is the leading contributor for floating-rate issuers, longer-maturity bonds, firm size, operating leverage, markup, and input-cost exposure. The floating-rate split is especially informative. The expected-inflation contribution is close to zero, while inflation uncertainty is the leading source of explained variation, matching the coefficient evidence. Inflation cyclicality is secondary in most dimensions, but becomes more visible among floating-rate issuers and longer-maturity bonds, where expected inflation contributes less.

The decomposition also shows why sensitivity and importance should be kept separate. Several operating and fundamentals sorts, including profitability, markup, operating leverage, and input-cost exposure, produce ordered coefficients but relatively flat explanatory-power profiles. Inflation risk therefore explains a similar share of residual variation across these groups even though the coefficients differ. These margins help identify the mechanism, but they are not where the explanatory power is most concentrated. The largest differences in contribution arise along credit rating, floating-rate exposure, debt growth, time to maturity, firm size, leverage, and refinancing intensity. These are the balance-sheet and debt-contract dimensions emphasized by the nominal-debt channel.

Taken together, the cross-sectional evidence shows that inflation risk is priced where it is most relevant for credit risk. Expected inflation is most important when nominal liabilities are large and slow to adjust. Inflation uncertainty is strongest when default risk is more exposed to future nominal conditions, especially for weaker firms, longer-maturity bonds, and firms with more vulnerable cash flows. Inflation cyclicality plays a smaller and less systematic role. The coefficients show where sensitivities are strongest, while the variance decomposition shows which inflation-risk component accounts for the explained variation. Both sets of results lead to the same conclusion. The explanatory power of inflation risk is concentrated among firms whose liabilities, cash flows, and debt contracts make credit risk more sensitive to nominal conditions.

V. Additional Evidence and Robustness

Having shown that inflation-risk proxies explain residual movements in corporate bond spreads, I now test whether the evidence is robust across measurement choices, specifications, residual constructions, and inflation regimes. I first vary the measurement of inflation risk beyond the swap and survey-anchored specifications studied above. The tests use fixed-maturity swaps, TIPS breakevens, CPI-based measures, and alternative cyclicality proxies to check whether the results depend on how inflation risk is mapped to corporate bond horizons. I then add controls for other aggregate forces that may move credit spreads together with inflation risk, including macroeconomic variables, monetary-policy measures, inflation volatility risk, firm-level equity volatility, and factors based on expected real growth. Finally, I revisit the residual common component by changing the cohort definitions and extending the analysis with the Enhanced TRACE sample through 2024. Across these exercises, the main results remain stable and are not tied to a single inflation proxy, residual grouping, or sample period.

A. *Alternative Measures of Inflation Risk*

The baseline analysis uses duration-matched zero-coupon inflation swaps to construct the inflation-risk proxies. This choice matches each corporate bond to the relevant point on the inflation curve and preserves the maturity-specific nature of the exposure. Section III.C has already shown that the results are similar when the inflation curve is anchored with survey expectations. I now examine whether the evidence is also robust to other ways of measuring inflation risk, including fixed-maturity swaps, TIPS breakevens, CPI-based proxies, alternative uncertainty measures, and alternative cyclicality measures.

Table V reports the results. Column (1) includes the full set of CDGM credit-risk controls and the intermediation controls. Column (2) repeats the baseline specification using duration-matched inflation swaps. Column (3) uses the 10-year inflation swap rate for all bonds, removing bond-specific duration matching while staying within the same inflation-swap market. Column (4) uses seasonally adjusted TIPS breakeven rates from Gürkaynak, Sack and Wright (2010), matched to bond duration. Column (5) uses CPI-based proxies constructed from realized inflation, following Boons, Duarte, de Roon and Szymanowska (2020). These CPI-based measures provide a useful check because they are based on realized inflation rather than inflation-linked security prices.

Table V about here

The signs are stable across these alternative constructions. The expected-inflation component remains negatively related to yield spread changes, while the uncertainty component remains positively related to yield spread changes. The explanatory power is similar across measures. The baseline swap specification explains 27.0% of residual variation, compared with 25.8% using the 10-year swap rate, 28.8% using TIPS breakevens, and 20.4% using CPI-based proxies. The CPI specification is weaker, as expected, because realized inflation is more backward-looking, but it delivers the same qualitative conclusion.

I also control directly for lagged realized inflation. I do not include contemporaneous CPI inflation because it is released with a delay and is not in the investor information set when prices are measured. Lagged CPI inflation, however, captures recently observed inflation conditions that may affect spreads through backward-looking updating or through macroeconomic conditions correlated with inflation realizations. Including lagged CPI inflation leaves the main results essentially unchanged.

I next examine alternative measures of inflation cyclicality. The baseline cyclicality proxy is the rolling correlation between changes in expected inflation and revisions in 12-month-ahead expected real GDP growth from Consensus Economics. This measure is designed to capture whether inflation news arrives together with stronger or weaker expected real activity. As a reduced-form alternative, I use the stock-bond return correlation, which is closely related to the cyclicality of inflation risk and the hedge value of nominal bonds. I compute this measure as the three-month rolling correlation between the 10-year Treasury bond return and the S&P 500 return, and define $\Delta\rho_t^{SB}$ as its monthly change. A higher stock-bond correlation indicates that nominal bonds provide less hedge value in bad real states, consistent with a greater role for inflation shocks that are adverse for real activity. I therefore expect increases in this proxy to be associated with wider credit spreads.

Finally, I examine whether the baseline uncertainty proxy is related to disagreement among forecasters. The baseline measure, $\Delta\sigma_{i,t}^{S}$, is based on time-series variation in the inflation swap curve at the bond's horizon. Forecast disagreement captures a different concept because it measures dispersion in beliefs across forecasters. To compare the two measures, I augment the baseline specification with changes in Consensus Economics forecast disagreement. The results suggest that disagreement and swap volatility capture different dimensions of inflation uncertainty. Forecast disagreement enters with the expected positive sign, although the estimate is only marginally significant. More importantly, the inflation-swap volatility proxy remains positive after controlling for disagreement. The same conclusion holds when I replace swap volatility with the component orthogonal to forecast disagreement. The baseline uncertainty result is therefore not driven only by cross-sectional disagreement among forecasters but also reflects time-series instability in the inflation outlook that matters for corporate bond spreads.

Overall, the results show that the main findings are not tied to one inflation-risk measure and remain robust to controls for realized inflation and forecast disagreement.

B. *Macro Conditions and the Residual Common Factor*

The next set of tests asks whether the inflation-risk measures are capturing variation that is better described by broader macroeconomic or monetary-policy conditions. This is especially relevant for the common-factor evidence. The first principal component of residual yield spread changes is a latent object, and inflation risk moves together with real activity, monetary policy, and aggregate risk. The question is therefore not whether inflation is correlated with these forces, but whether it explains residual spread variation after they are accounted for directly.

Table VI starts from the bond-level specification and adds variables that are natural alternatives to the inflation-risk interpretation. Column (1) reports the full baseline specification. The following columns add additional aggregate controls one at a time. I first include the inflation volatility risk factor of Ceballos (2021); I then add real consumption growth, income growth, and changes in unemployment, which capture broad variation in aggregate credit conditions. Finally, I include monetary-policy variables such as changes in the federal funds rate and the monetary-policy uncertainty measure of Bu, Rogers and Wu (2021). The inflation-risk coefficients remain stable across these specifications, and the explanatory power also remains close to the baseline. Thus, the inflation-risk measures do not appear to be capturing only standard macroeconomic variation or monetary-policy movements.

Table VI about here

The same table also adds firm-level equity volatility, following Campbell and Taksler (2003). This is an important control because equity volatility is a direct measure of firm-level risk and is known to be strongly related to corporate bond spreads. Equity volatility enters with the expected positive sign, confirming that firm-level volatility matters for yield spread changes. Adding it does not eliminate the inflation-risk effects, indicating that the results are not only a restatement of the standard volatility channel in corporate credit.

After the bond-level controls in Table VI, I turn to the residual common component. Table VII compares inflation risk with macroeconomic and monetary-policy controls, growth-expectation variables, and inflation-risk proxies orthogonalized to standard aggregate conditions.

I next compare inflation risk with a set of macroeconomic, monetary-policy, and growth-expectation variables. Expected inflation and expected real activity are related, but they are not the same object. Growth forecasts capture expected real cash-flow conditions, while inflation risk also captures changes in the real value of nominal liabilities and uncertainty about the future price level. Table VII therefore reports the comparison in separate specifications. With the CDGM controls, the inflation-risk block explains 36.6% of residual variation and 22.3% of the variance of PC1. The macro, monetary-policy, and growth-expectation variables explain 43.3% of residual variation and 24.6% of PC1 variance. With the full set of controls, the corresponding numbers are 27.0% and 7.2% for inflation risk, and 31.7% and 8.7% for the macro, monetary-policy, and growth-expectation block. These separate regressions show that expected real activity and aggregate conditions are also closely related to the residual common component.

Table VII about here

I then orthogonalize the inflation-risk proxies with respect to the set of macro variables. This provides a sharper test of whether inflation risk explains residual spread variation beyond its relation with standard aggregate conditions.

The orthogonalized inflation-risk proxies continue to explain residual variation, even after removing their relation with macroeconomic, monetary-policy, and growth-expectation variables. Their explanatory power is lower than in the baseline specification, as expected, but remains economically meaningful: 18.8% of residual variation with the CDGM controls and 20.0% with the full set of credit-risk and intermediation controls.

I then combine the orthogonalized inflation-risk proxies with the macroeconomic, monetary-policy, and growth-expectation variables in a single horse-race specification. The combined model explains 54.6% of residual variation with the CDGM controls and 44.9% with the full controls, with corresponding PC1 values of 31.7% and 13.1%. The increase relative to either set of variables alone shows that inflation risk does not exhaust the residual common component, but explains a separate part of it.

C. *Alternative Residual Groupings and Extended Sample*

Because the common factor is extracted from residuals averaged within maturity-leverage portfolios, I test whether the results are sensitive to the CDGM grouping choice.

Table VIII repeats the exercise using alternative residual groupings. I keep maturity as one sorting dimension and pair it with leverage, rating, trading volume, market beta, and VIX beta. For each grouping, I compute cohort-month average residuals and extract principal components from the covariance matrix. Across these specifications, inflation risk explains between 23.9% and 29.1% of residual variation. This range is close to the 27.0% estimate from the baseline maturity-leverage grouping.

Table VIII about here

The effect on the residual common component is also stable. Without inflation risk, PC1 explains between 60.5% and 74.9% of residual variation, depending on the grouping. After adding the inflation-risk proxies, the PC1 share falls in every case, by about five percentage points. The decline is similar for maturity-leverage, maturity-rating, maturity-volume, maturity-market-beta, and maturity-VIX-beta portfolios. The common-factor evidence is therefore not driven by the specific way in which the residual portfolios are formed.

I next extend the sample through December 2024 using Enhanced TRACE. This extension adds the post-pandemic inflation surge and the Federal Reserve's rapid tightening cycle, a period in which inflation risk became much more salient for firms and financial markets. The post-2021 analysis is best viewed as a robustness check rather than a like-for-like extension of the baseline, because Enhanced TRACE lacks the dealer identifiers needed to construct the full set of OTC market proxies, so the extended specifications use a restricted intermediation-control set.[31]

Table IX reports the extended-sample results. The main signs remain unchanged and the explanatory power remains close to the baseline. In the fully controlled extended sample, the inflation-risk block explains 23.3% of residual variation, compared with 27.0% in the baseline sample. The PC1 regressions also continue to show that inflation risk is related to the common residual component, with a PC1 R^2 of 10.6%. Panel D reaches the same conclusion in fixed-effects regressions, where the signs of all three inflation-risk proxies are unchanged when the model is estimated on the extended panel.

Table IX about here

D. *Time Variation in Inflation Sensitivity*

I now examine whether the average relation between inflation risk and yield spread changes varies over time, since inflation should matter more for credit pricing when expectations move sharply, uncertainty rises, or inflation news becomes more informative about expected real activity.

Figure 5 plots 24-month rolling estimates of the coefficients on the three inflation-risk proxies. The expected-inflation coefficient is close to zero before the Global Financial Crisis, turns sharply negative during the crisis, and remains mostly negative afterward. The effect becomes especially large around the post-COVID inflation surge. This pattern is consistent with the nominal-debt channel, where declines in expected inflation widen spreads when they raise the real burden of outstanding nominal liabilities, and the effect becomes stronger when inflation risk is more salient.

Figure 5 about here

The uncertainty coefficient is less persistent. It is often close to zero in calmer periods, but becomes larger when inflation is more unstable, especially around the post-COVID episode. This pattern is consistent with the default-option channel. A less predictable inflation outlook increases uncertainty about nominal firm value, which can raise default

[31]The Enhanced TRACE database does not include dealer identifiers. In the extended-sample tests, I use the subset of OTC market proxies that can be constructed without them: changes in dealer inventory, amount outstanding, matched trades, block trades, the TED spread, and aggregate ratings. These variables represent the majority of the effect documented in Friewald and Nagler (2019).

risk and widen credit spreads.

The cyclicality proxy behaves differently from the other two measures. Its rolling coefficient is positive during parts of the Global Financial Crisis and again around some later episodes, but it is also more volatile and less persistent. This is consistent with inflation cyclicality as a state-dependent dimension of inflation risk, since its effect depends on whether inflation news arrives with stronger or weaker expected real activity.

The episode estimates summarize these patterns more directly. Before the Global Financial Crisis, the expected-inflation channel was essentially absent. The uncertainty effect is also weak. This is the period in which inflation expectations were most stable, so the lack of a strong inflation-spread relation is consistent with the rolling estimates.

During the Global Financial Crisis, the expected-inflation coefficient became more negative, and the cyclicality channel turned positive, consistent with a period in which lower inflation expectations coincided with weaker expected growth and a higher real burden of nominal debt. The 2014–2015 oil-price disinflation episode shows similar signs, but the estimates are smaller, consistent with a more sector-specific shock.

The post-COVID period shows a different pattern. Expected inflation remains negatively related to spreads, while the uncertainty coefficient becomes larger, pointing to a more visible role for the uncertainty channel. The cyclicality estimate is less stable, consistent with an environment in which inflation news was harder to interpret because high inflation coincided with supply constraints, fiscal expansion, and rapid monetary tightening.

Overall, the time-series evidence helps interpret the pooled results. Expected inflation and inflation uncertainty have stable average signs, but their strength changes over time. The effects are muted when inflation is stable and become stronger when inflation risk is more salient. Cyclicality is more regime dependent, which explains why its pooled coefficient is weaker and less stable than the coefficients on expected inflation and inflation uncertainty.

VI. Conclusion

This paper studies whether inflation risk explains the part of corporate yield spread changes that standard credit-risk and intermediation variables leave unexplained. I find that expected inflation, inflation uncertainty, and inflation cyclicality account for a sizeable share of this residual variation, including its common component across bonds. Spreads narrow when expected inflation rises and widen when inflation uncertainty increases. These results suggest that inflation risk captures an important source of spread variation that is not absorbed by standard credit-risk and intermediation controls.

The evidence is consistent with nominal-debt models with slow leverage adjustment. In these models, expected inflation compresses spreads by reducing the real burden of promised payments, while inflation uncertainty widens spreads by increasing the value of the default option. The time-series evidence reinforces this interpretation. Inflation sensitivity is limited during stable inflation periods, but becomes stronger during episodes of macroeconomic stress, especially after COVID.

The cross-sectional evidence points to the same interpretation. Inflation risk matters most where credit risk is more exposed to nominal debt and inflationary cost pressures, including highly levered firms, firms with weaker fundamentals, firms whose debt terms adjust more slowly, and firms with less pricing power or greater input-cost exposure. Expected inflation is more important when nominal liabilities are large or slow to adjust, while inflation uncertainty is more relevant when default risk, maturity, liability repricing, or operating exposure make firms more sensitive to future nominal conditions. Inflation cyclicality plays a smaller role overall, but becomes more visible when the expected-inflation channel is muted, consistent with its state-dependent interpretation. Overall, the evidence shows that inflation risk matters most where firm balance sheets, cash flows, and debt contracts make credit risk more sensitive to inflation.

These findings suggest several avenues for future research. One natural direction is to examine whether the heterogeneous effects documented here extend to other asset classes.

For example, the interaction between inflation risk and refinancing needs could be studied in equity prices by building on models such as Bhamra, Dorion, Jeanneret and Weber (2022) and allowing for a richer debt maturity structure. A second avenue is to study firms' financing and issuance decisions under different inflation scenarios. If inflation changes the real value of nominal liabilities and the cost of refinancing, firms may adjust debt maturity, coupon structure, floating-rate exposure, or issuance timing in ways that are not captured by spread changes alone. This would extend the financing implications of the nominal-debt channel studied in Gomes, Jermann and Schmid (2016) and Gomes and Schmid (2020).

In conclusion, inflation risk accounts for a significant share of the systematic residual variation in yield spread changes and sheds light on the common component that standard credit-risk variables fail to capture. The results show that this common component is systematically related to nominal conditions and to the channels through which inflation affects credit risk. This evidence supports structural models that combine nominal debt, slow leverage adjustment, and inflation dynamics. More broadly, it suggests that inflation risk is an important state variable for corporate credit markets, especially when changes in nominal conditions alter the real value and riskiness of promised payments.

Figure 1: **Heterogeneity in Inflation Risk: Firm Fundamentals and Credit Quality**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). Bonds are assigned to groups based on the average value of the relevant characteristic over the bond's sample life. The figure reports average estimated coefficients within each group, with 95% confidence intervals shown by bars and median coefficients shown by dots. Columns report results for size, return on assets, profit margin, and credit rating groups. Rows correspond to expected inflation, inflation uncertainty, and inflation cyclicality. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

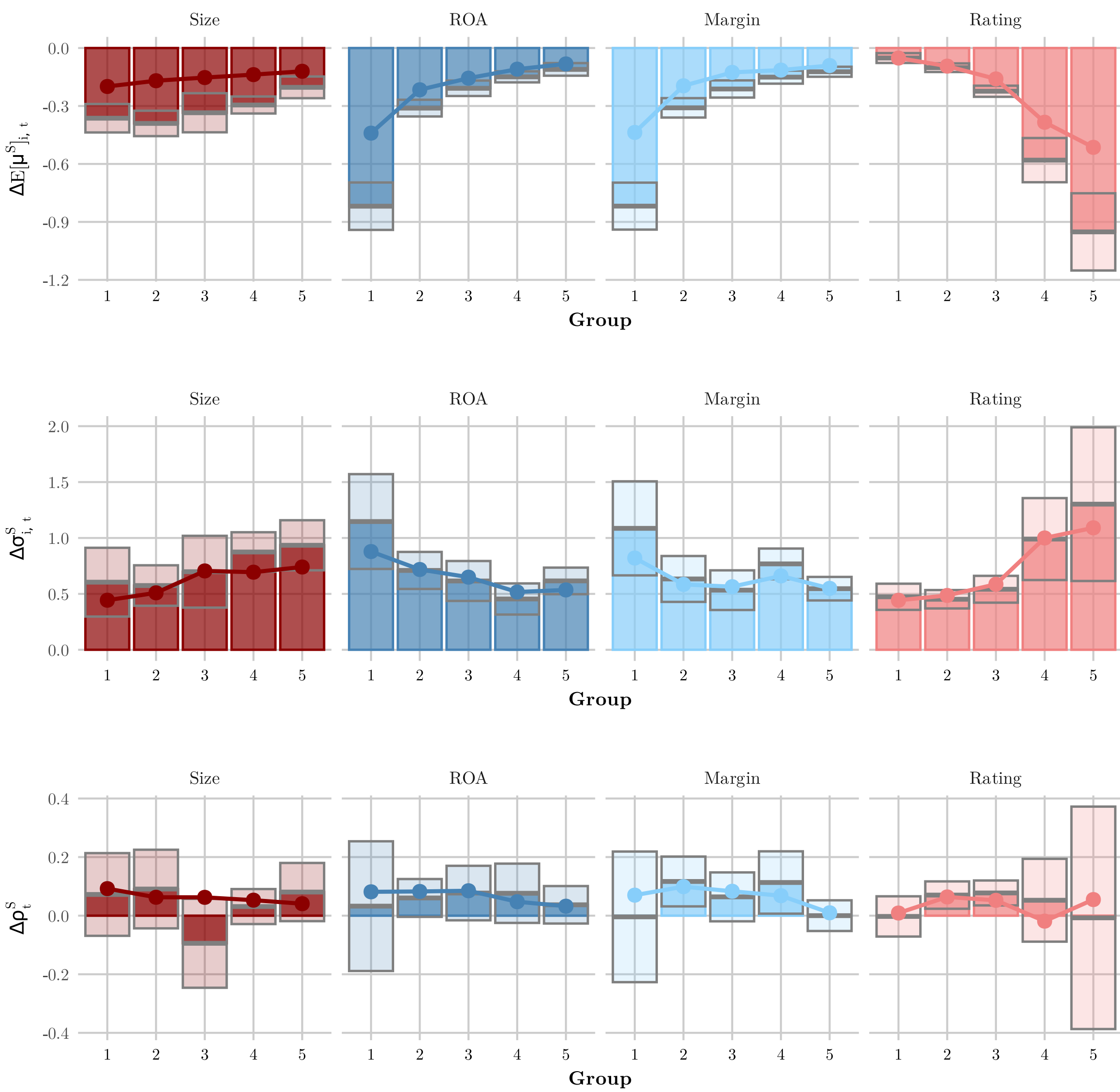


Figure 2: **Heterogeneity in Inflation Risk: Operating Channels**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). Bonds are assigned to groups based on the average value of the relevant characteristic over the bond's sample life. The figure reports average estimated coefficients within each group, with 95% confidence intervals shown by bars and median coefficients shown by dots. Columns report results for operating leverage, markup, and input-cost exposure. Rows correspond to expected inflation, inflation uncertainty, and inflation cyclicality. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

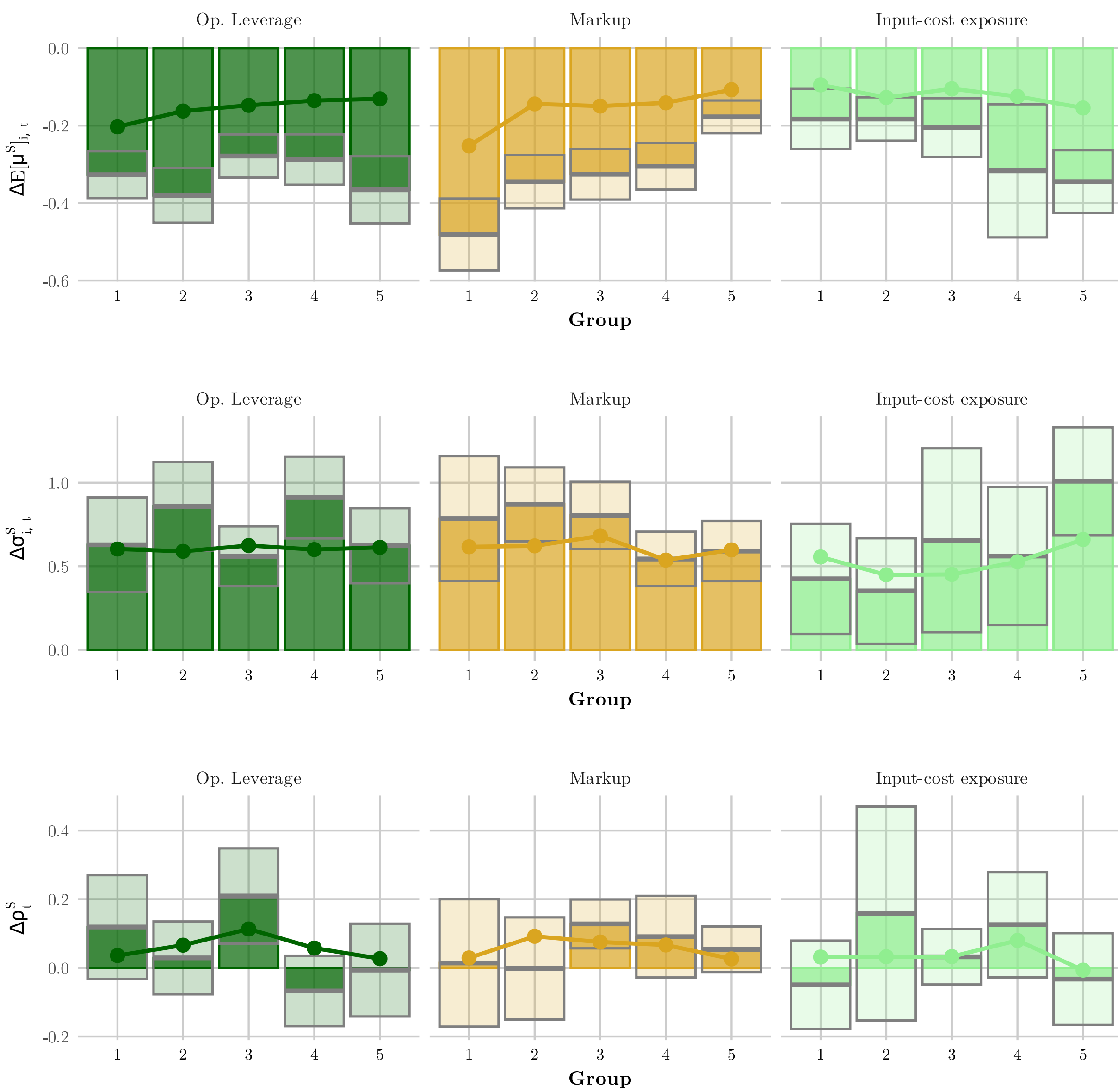


Figure 3: **Heterogeneity in Inflation Risk: Debt Structure**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). Bonds are assigned to groups based on the average value of the relevant characteristic over the bond's sample life. The figure reports average estimated coefficients within each group, with 95% confidence intervals shown by bars and median coefficients shown by dots. Columns report results for leverage, refinancing intensity, debt growth, floating-rate exposure and maturity. Rows correspond to expected inflation, inflation uncertainty, and inflation cyclicality. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

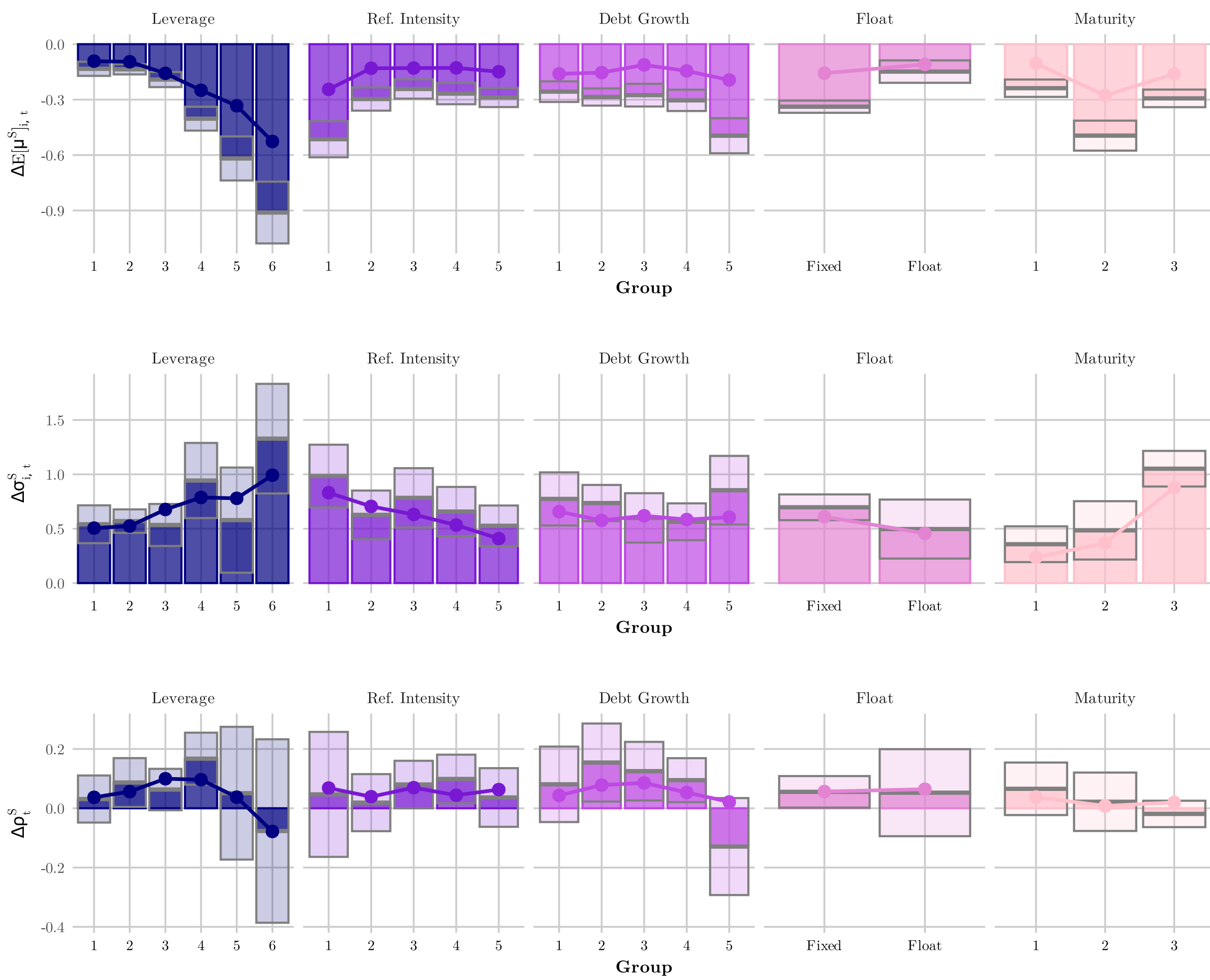


Figure 4: **Which Dimension of Inflation Risk Explains Residual Spread Variation**

For each heterogeneity dimension, bonds are assigned to groups using the average value of the relevant characteristic over the bond's sample life. I first remove the variation explained by the same structural variables and controls used in the coefficient regressions. I then compute the FVE from adding expected inflation, inflation uncertainty, and inflation cyclicality to the control specification.

Each proxy is added separately, and contributions are measured relative to a common residual-variance denominator within the same dimension. This makes the bars comparable across groups and makes group contributions sum to the dimension total. Stacked bars report the single-proxy contributions, with expected inflation in red, inflation uncertainty in dark blue, and inflation cyclicality in light blue. Diamonds report the joint FVE from adding all three proxies together. Since the proxies are correlated, the distance between the stacked bars and the diamonds reflects shared explanatory power. Within each panel, groups are ordered from low to high. Panels are ordered by the cross-group variation in joint FVE. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

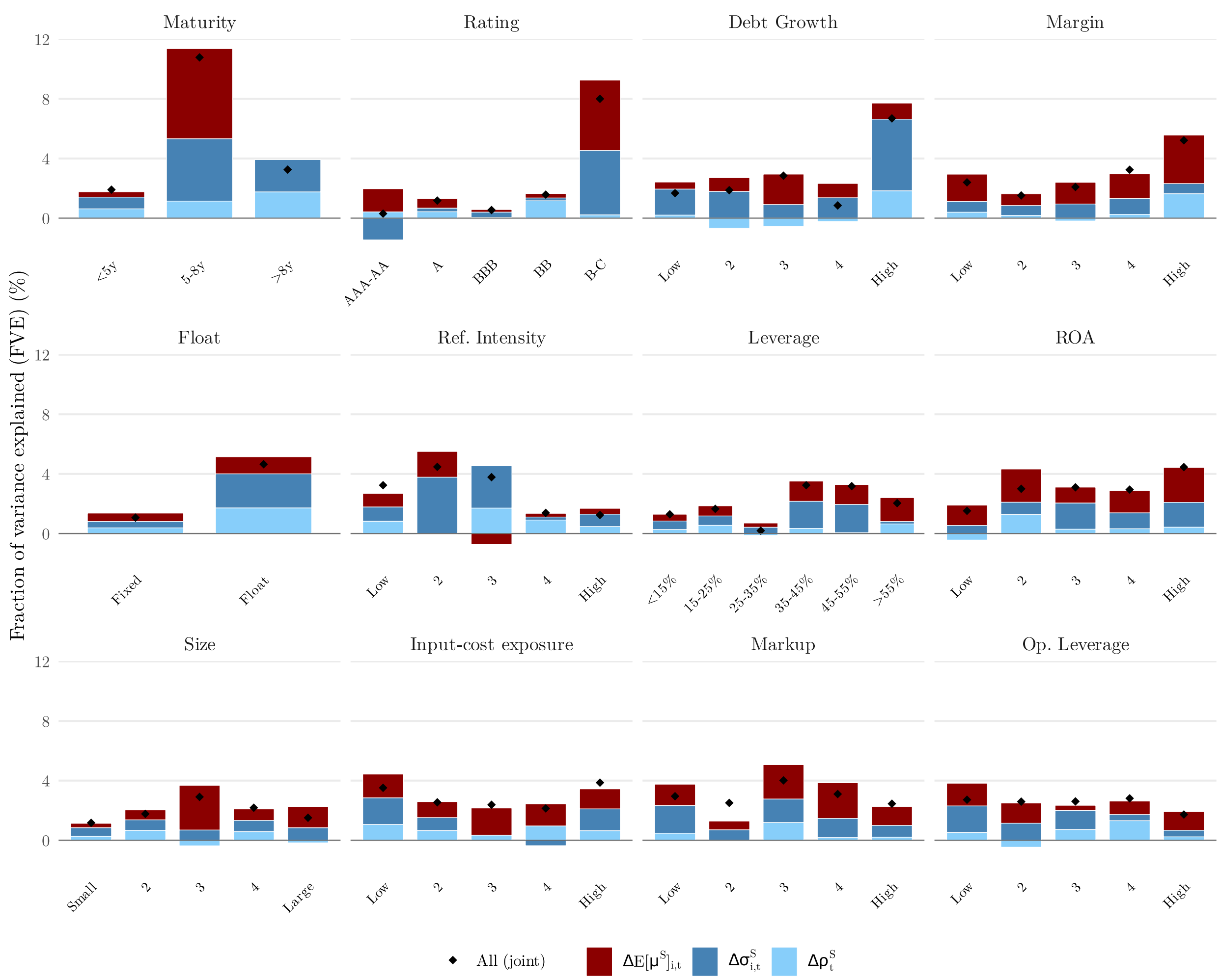


Figure 5: **Rolling Estimates of Inflation Risk Sensitivities**

This figure reports rolling estimates of the sensitivity of yield spread changes to the proxies of inflation risk. For each 24-month window, I estimate the baseline model separately for each bond observed in that window and then average the estimated coefficients across bonds in that window. The model includes the CDGM structural credit-risk controls and the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III, but excludes intermediation controls. The left panel reports expected inflation, the middle panel reports inflation uncertainty, and the right panel reports inflation cyclicality. Grey bands show 95% confidence intervals. Shaded regions denote the Global Financial Crisis, the oil-price disinflation episode, and the post-COVID inflation surge. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

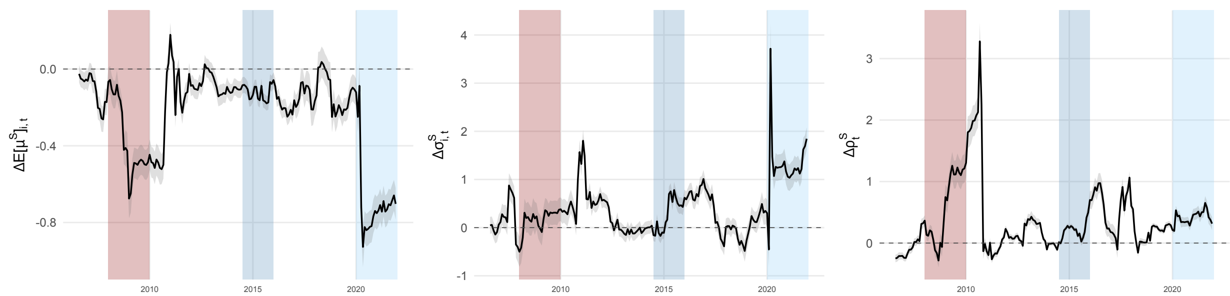


Table I: **Summary Statistics**

This table reports summary statistics of the data. Panel A reports the number of observations, mean, standard deviation, and 5%, 25%, 50%, 75% and 95% quantiles of bond characteristics, yield spreads ($YS_{i,t}$), and the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III. The bond characteristics include the offering amount, the coupon rate, the bond age, the time to maturity, the duration, and the credit rating. The bond's rating is determined as the average of ratings provided by Standard & Poor's (S&P), Moody's, and Fitch when more than one are available or as the rating provided by one of the three rating agencies when only one rating is available. Panel B reports the standard deviation and the correlation matrix of changes in yield spreads and changes in the proxies of inflation risk. The sample is based on U.S. corporate bond transaction data from TRACE for the period 2004–2021.

Panel A: Summary Statistics

	Obs.	Mean	Std.	5%	25%	50%	75%	95%
Offering amount (mil.)	449788	741.84	658.33	200	350	500	1,000	2,000
Coupon (%)	449788	5.39	1.97	2.25	3.88	5.38	6.88	8.62
Age	449788	5.51	5.33	0.52	2.05	3.98	7.06	17.43
Time to Maturity	449788	8.99	8.08	0.96	3.38	6.15	10.00	27.27
Duration	449788	6.49	4.29	1.00	3.32	5.46	8.25	15.54
Rating	449788	8.75	3.47	4	6	9	10	15
Monthly Volume (bil.)	449788	5.37	10.09	0.07	0.71	2.42	6.26	19.83
Leverage	449788	0.31	0.21	0.07	0.16	0.26	0.42	0.75
Yield spread (%)	449788	2.32	3.05	0.32	0.81	1.44	2.71	6.57
$\Delta YS_{i,t}$	449788	0.74	82.73	-64.79	-14.89	-1.06	12.64	67.38
$E[\mu^S]_{i,t}$ (%)	449788	2.09	0.59	1.20	1.79	2.13	2.48	2.89
$\Delta E[\mu^S]_{i,t}$	449788	-0.18	25.10	-30.99	-9.62	0.59	11.61	30.67
$\Delta\sigma^S_{i,t}$	449788	0.07	7.27	-6.83	-2.04	-0.07	1.91	7.70
$\Delta\rho^S_t$	449788	-0.05	10.12	-14.31	-2.35	0.32	3.78	11.83

Panel B: Correlation Matrix

	Std.	$\Delta YS_{i,t}$	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$
$\Delta YS_{i,t}$	0.827	1	-0.299	0.236	0.167
$\Delta E[\mu^S]_{i,t}$	0.251		1	-0.346	-0.154
$\Delta\sigma^S_{i,t}$	0.073			1	0.355
$\Delta\rho^S_t$	0.101				1

Table II: **Inflation Risk and Yield Spread Changes**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma_{i,t}^S$, $\Delta\rho_t^S$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). Column (1) reports the CDGM regression; Columns (2)–(4) include each proxy individually; Column (5) includes all three jointly; Columns (6)–(8) progressively add controls. Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity × 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variance explained (FVE) by adding the inflation-risk proxies. Panel C includes statistics from time-series regressions of the PC1 on the inflation-risk proxies. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Individual Bond Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.537			-0.480	-0.382	-0.393	-0.328
		(-37.623)			(-35.821)	(-25.272)	(-24.936)	(-20.979)
$\Delta\sigma_{i,t}^S$			1.600		1.279	1.124	1.165	0.716
			(36.336)		(30.383)	(24.245)	(21.798)	(12.873)
$\Delta\rho_t^S$				0.604	0.252	0.077	0.062	0.056
				(31.403)	(14.981)	(3.048)	(2.265)	(1.912)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Mean R^2	0.355	0.386	0.418	0.390	0.447	0.507	0.516	0.530
Median R^2	0.378	0.410	0.431	0.410	0.462	0.541	0.554	0.573
Obs.	449788	449788	449788	449788	449788	449788	449788	449788
Bonds	6826	6826	6826	6826	6826	6826	6826	6826
Panel B: Principal Component Analysis								
FVE		0.200	0.179	0.088	0.366	0.288	0.287	0.270
PC1	0.797	0.743	0.759	0.786	0.712	0.691	0.686	0.653
PC2	0.054	0.078	0.064	0.059	0.084	0.084	0.092	0.097
UV	1.123	0.899	0.922	1.025	0.712	0.441	0.392	0.318
Panel C: Time-Series Regression of PC1 on Inflation-Risk Proxies								
Adj. R^2		0.129	0.152	0.062	0.212	0.114	0.099	0.058
R^2		0.133	0.157	0.067	0.223	0.126	0.112	0.072
F-stat		31.565	38.231	14.783	19.545	9.838	8.543	5.279
p-value		0.000	0.000	0.000	0.000	0.000	0.000	0.002
Obs.		208	208	208	208	208	208	208

Table III: **Inflation Expectations and Yield Spread Changes**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ is the vector of CDGM structural credit-risk controls, $\Delta I_{i,t}$ contains survey-anchored proxies of inflation risk ($\Delta E[\mu^E]_{i,t}$, $\Delta\sigma^E_{i,t}$, $\Delta\rho^E_t$), and $\Delta C_{i,t}$ contains the intermediation controls from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). The survey-anchored proxies are obtained from a Nelson-Siegel-Svensson curve fitted daily to inflation swaps, TIPS breakevens, and survey expectations, and evaluated at each bond's duration. Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity $\times$ 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variance explained (FVE) by adding the inflation-risk proxies. Panel C includes statistics from time-series regressions of the PC1 on the inflation-risk proxies. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Individual Bond Regressions						
$\Delta E[\mu^E]_{i,t}$		-0.260		-0.300		-0.283
		(-19.326)		(-21.198)		(-19.259)
$\Delta\sigma^E_{i,t}$			0.504	0.407		0.378
			(9.558)	(6.521)		(5.401)
$\Delta\rho^E_t$					0.173	0.171
					(3.119)	(2.629)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes
FN	Yes	Yes	Yes	Yes	Yes	Yes
HKS	Yes	Yes	Yes	Yes	Yes	Yes
EHL	Yes	Yes	Yes	Yes	Yes	Yes
Mean R^2	0.505	0.515	0.519	0.528	0.511	0.532
Median R^2	0.534	0.546	0.548	0.560	0.544	0.570
Obs.	449788	449788	449788	449788	449788	449788
Bonds	6826	6826	6826	6826	6826	6826
Panel B: Principal Component Analysis						
FVE		0.100	0.086	0.172	0.119	0.246
PC1	0.708	0.677	0.696	0.667	0.683	0.653
PC2	0.085	0.097	0.086	0.100	0.091	0.103
UV	0.436	0.393	0.399	0.361	0.385	0.329
Panel C: Time-Series Regression of PC1 on Inflation-Risk Proxies						
Adj. R^2		0.039	0.033	0.047	0.009	0.044
R^2		0.044	0.038	0.057	0.014	0.057
F-stat		9.477	8.132	6.140	2.871	4.139
p-value		0.002	0.005	0.003	0.092	0.007
Obs.		208	208	208	208	208

Table IV: **Inflation Risk and Yield Spread Changes: Estimation Approach and Inference**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the fully controlled specification including the CDGM structural credit-risk controls, the intermediation controls from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024), and the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III. Column (1) reports mean-group estimates, obtained by estimating the model separately for each bond and averaging coefficients across bonds. The t-statistics in column (1) are computed from the cross-sectional dispersion of bond-level coefficient estimates, following CDGM. Columns (2)–(4) report panel fixed-effects estimates with bond fixed effects and homogeneous slopes. The columns differ only in the clustering of standard errors: column (2) double-clusters by bond and month, column (3) clusters by month, and column (4) clusters by bond. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	Individual Bond Regressions	Panel FE		
	(CDGM)	Bond×Time	Time only	Bond only
	(1)	(2)	(3)	(4)
$\Delta E[\mu^S]_{i,t}$	-0.328***	-0.340***	-0.340***	-0.340***
	(-20.98)	(-4.54)	(-4.54)	(-19.05)
$\Delta\sigma^S_{i,t}$	0.716***	0.587***	0.587***	0.587***
	(12.87)	(2.86)	(2.90)	(8.42)
$\Delta\rho^S_t$	0.056*	0.166	0.166	0.166***
	(1.91)	(1.40)	(1.40)	(9.96)
CDGM	Yes	Yes	Yes	Yes
FN	Yes	Yes	Yes	Yes
HKS	Yes	Yes	Yes	Yes
EHL	Yes	Yes	Yes	Yes
Bond FE	Bond-level TS	Yes	Yes	Yes
Estimator	Mean Group	Panel FE	Panel FE	Panel FE
Clustering	Cross-bond	Bond×Time	Time only	Bond only
Observations	449788	449788	449788	449788

Table V: **Alternative Inflation-Risk Proxies**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ contains the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). All columns include $\Delta S_{i,t}$ and $\Delta C_{i,t}$. Column (1) excludes inflation-risk proxies. Columns (2)–(6) use alternative proxy sets: baseline duration-matched swaps, 10-year swaps, duration-matched TIPS breakevens, CPI-based proxies, and the stock–bond cyclicality proxy. Columns (7)–(10) add, respectively, lagged CPI inflation, lagged CPI inflation with the baseline swap proxies, forecast disagreement, and inflation uncertainty orthogonalized to forecast disagreement. Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity × 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variance explained (FVE) by adding the inflation-risk proxies. Panel C includes statistics from time-series regressions of the PC1 on the inflation-risk proxies. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Individual Bond Regressions										
Proxies		Swaps	Swaps 10Y	TIPS	CPI		Swaps			Swaps orth.
$\Delta E[\mu]_{i,t}$		-0.328	-0.329	-0.314	-0.123	-0.352		-0.365	-0.313	-0.309
		(-20.979)	(-13.610)	(-21.760)	(-11.655)	(-22.375)		(-13.602)	(-17.452)	(-19.922)
$\Delta\sigma_t$		0.716	0.886	0.415	0.242	0.700		0.748	0.803	0.707
		(12.873)	(9.359)	(7.559)	(6.407)	(11.001)		(12.306)	(10.447)	(12.458)
$\Delta\rho_t$		0.056	0.020	0.204	0.172			0.102	0.101	0.089
		(1.912)	(0.740)	(5.837)	(3.327)			(3.369)	(2.142)	(3.009)
$\Delta\rho^{SB}_t$						0.106				
						(5.387)				
CPI lag							7.030	5.952		
							(10.181)	(6.532)		
CE disag.									0.172	
									(2.612)	
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HKS	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
EHL	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean R^2	0.505	0.530	0.525	0.529	0.512	0.527	0.509	0.533	0.534	0.530
Median R^2	0.534	0.573	0.570	0.570	0.551	0.564	0.541	0.582	0.577	0.571
Obs.	449788	449788	449788	449788	449788	449788	449788	449788	449788	449788
Bonds	6826	6826	6826	6826	6826	6826	6826	6826	6826	6826
Panel B: Principal Component Analysis										
FVE		0.270	0.258	0.288	0.204	0.263	0.074	0.298	0.303	0.274
PC1	0.708	0.653	0.654	0.644	0.669	0.655	0.695	0.650	0.651	0.652
PC2	0.085	0.097	0.095	0.100	0.096	0.098	0.087	0.097	0.096	0.095
UV	0.436	0.318	0.324	0.311	0.347	0.322	0.404	0.306	0.304	0.317
Panel C: Time-Series Regression of PC1 on Inflation-Risk Proxies										
Adj. R^2		0.058	0.095	0.107	0.060	0.068	0.027	0.084	0.066	0.062
R^2		0.072	0.108	0.119	0.074	0.081	0.032	0.102	0.084	0.075
F-stat		5.279	8.237	9.225	5.431	6.023	6.805	5.733	4.669	5.543
p-value		0.002	0.000	0.000	0.001	0.001	0.010	0.000	0.001	0.001
Obs.		208	208	208	208	208	208	208	208	208

Table VI: **Additional Variables**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ contains the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). Odd-numbered columns add auxiliary controls only, while even-numbered columns also add the inflation-risk proxies. The auxiliary controls are inflation volatility risk in Columns (1)–(2), consumption growth, income growth, and unemployment in Columns (3)–(4), monetary-policy uncertainty and the federal funds rate in Columns (5)–(6), and firm-level equity volatility in Columns (7)–(8). Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity × 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variance explained (FVE) by adding the inflation-risk proxies. Panel C reports statistics from time-series regressions of PC1 on the variables added in each specification. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Individual Bond Regressions								
ΔIVR_t	0.293	0.235						
	(8.830)	(4.245)						
$\Delta Consumption_t$			3.212	2.851				
			(2.961)	(2.669)				
$\Delta Income_t$			3.076	2.947				
			(5.467)	(4.112)				
$\Delta Unemployment_t$			0.088	0.058				
			(4.446)	(3.080)				
ΔMPU_t					-0.004	0.000		
					(-0.749)	(0.050)		
$\Delta FED\ Fund_t$					0.003	0.106		
					(0.091)	(2.673)		
$\Delta\sigma^{Equity}_{i,t}$							0.602	0.283
							(6.015)	(2.295)
$\Delta E[\mu^S]_{i,t}$		-0.322		-0.217		-0.321		-0.314
		(-16.598)		(-12.052)		(-18.999)		(-16.247)
$\Delta\sigma^S_{i,t}$		0.654		0.791		0.801		0.744
		(9.947)		(9.714)		(11.263)		(10.685)
$\Delta\rho^S_t$		0.099		0.079		0.055		0.054
		(2.657)		(2.679)		(1.875)		(1.592)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HKS	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
EHL	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean R^2	0.505	0.530	0.524	0.542	0.511	0.536	0.506	0.529
Median R^2	0.537	0.578	0.559	0.585	0.541	0.577	0.536	0.575
Obs.	449788	449788	449788	449788	449788	449788	449788	449788
Bonds	6826	6826	6826	6826	6826	6826	6826	6826
Panel B: Principal Component Analysis								
FVE		0.271		0.256		0.281	0.059	0.305
PC1	0.705	0.649	0.687	0.631	0.706	0.645	0.707	0.654
PC2	0.087	0.099	0.081	0.097	0.083	0.097	0.083	0.096
UV	0.414	0.302	0.351	0.261	0.407	0.293	0.411	0.303
Panel C: Time-Series Regression of PC1 on Inflation-Risk Proxies								
Adj. R^2		0.061		0.055		0.057	0.007	0.055
R^2		0.074		0.068		0.071	0.011	0.073
F-stat		5.445		4.987		5.171	2.385	3.984
p-value		0.001		0.002		0.002	0.124	0.004
Obs.		208		208		208	208	208

Table VII: **Macro Conditions and the Residual Common Component**

This table compares the explanatory power of inflation risk with macroeconomic, monetary-policy, and growth-expectation variables for the residual common component in corporate yield spread changes. The residuals are formed from bond-level spread-change regressions and averaged within maturity–leverage cohorts. The table reports results under two benchmark specifications: the CDGM structural credit-risk controls and the fully controlled specification that also includes the intermediation controls from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). Columns compare four regressor blocks: the baseline inflation-risk proxies, macroeconomic, monetary-policy, and growth-expectation variables, inflation-risk proxies orthogonalized with respect to aggregate conditions, and a horse-race specification combining the orthogonalized inflation-risk proxies with the macroeconomic, monetary-policy, and growth-expectation variables. Panel A reports principal component statistics for the cohort residuals, including PC1, PC2, total unexplained variance (UV), and the fraction of variation explained (FVE) by adding the relevant block. Panel B reports statistics from time-series regressions of the residual PC1 on the corresponding block variables. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	Inflation (1)	Macro+MP+Growth (2)	Infl.$^{\perp}$ (3)	Horse race (4)
CDGM				
Panel A: Principal Component Analysis				
FVE	0.366	0.433	0.188	0.546
PC1	0.712	0.718	0.771	0.683
PC2	0.084	0.073	0.058	0.080
UV	0.712	0.637	0.912	0.509
Panel B: Time-Series Regression of PC1 on Inflation-Risk or Macro Proxies				
Adj. R^2	0.212	0.224	0.058	0.286
R^2	0.223	0.246	0.071	0.318
F-stat	19.545	10.944	5.217	10.235
p-value	0.000	0.000	0.002	0.000
Obs.	208	208	208	208
CDGM + FN+HKS+EHL				
Panel A: Principal Component Analysis				
FVE	0.270	0.317	0.200	0.449
PC1	0.653	0.672	0.679	0.629
PC2	0.097	0.086	0.090	0.096
UV	0.318	0.298	0.349	0.241
Panel B: Time-Series Regression of PC1 on Inflation-Risk or Macro Proxies				
Adj. R^2	0.058	0.060	0.029	0.091
R^2	0.072	0.087	0.044	0.131
F-stat	5.279	3.206	3.093	3.312
p-value	0.002	0.005	0.028	0.001
Obs.	208	208	208	208

Table VIII: **Different Residual Groups**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model:

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ contains the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma_{i,t}^S$, $\Delta\rho_t^S$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). I then assign monthly residuals to cohorts using maturity as the first sorting variable and, alternatively, leverage, credit rating, trading volume, market beta, or VIX beta as the second sorting variable. For each grouping, I compute average cohort-month residuals and extract principal components from their covariance matrix. The table reports the share of residual variation explained by the first and second principal components (PC1 and PC2), separately for specifications without and with the inflation-risk proxies, and the fraction of variation explained (FVE) by adding the inflation-risk proxies. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

Group 1	Group 2	Inflation Risk	PC1	PC2	FVE
Time to Maturity	Leverage	No	0.708	0.085	
		Yes	0.653	0.097	0.270
Time to Maturity	Rating	No	0.659	0.113	
		Yes	0.605	0.125	0.269
Time to Maturity	Volume	No	0.749	0.075	
		Yes	0.697	0.089	0.291
Time to Maturity	Market Beta	No	0.640	0.090	
		Yes	0.591	0.099	0.251
Time to Maturity	VIX Beta	No	0.604	0.105	
		Yes	0.551	0.116	0.239

Table IX: **Extended Sample**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ is the vector of CDGM structural credit-risk controls, $\Delta I_{i,t}$ contains the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma_{i,t}^S$, $\Delta\rho_t^S$) introduced in Section III, and $\Delta C_{i,t}$ contains the available intermediation controls. Column (1) excludes the inflation-risk proxies. Columns (2)–(4) include each proxy individually. Column (5) includes all three jointly. Columns (6)–(8) progressively add the available controls. Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity × 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variance explained (FVE) by adding the inflation-risk proxies. Panel C includes statistics from time-series regressions of the PC1 on the inflation-risk proxies. Panel D reports panel fixed-effects estimates with standard errors double-clustered by bond and month. The sample is based on U.S. corporate bond transaction data from Enhanced TRACE for the period 2004–2024.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Individual Bond Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.337			-0.302	-0.284	-0.269	-0.229
		(-33.924)			(-32.055)	(-29.020)	(-27.357)	(-23.592)
$\Delta\sigma_{i,t}^S$			1.362		1.120	0.994	0.913	0.513
			(40.397)		(35.343)	(31.861)	(29.174)	(16.235)
$\Delta\rho_t^S$				0.461	0.242	0.219	0.209	0.131
				(29.469)	(17.226)	(14.490)	(13.588)	(8.721)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Mean R^2	0.323	0.343	0.373	0.345	0.394	0.436	0.446	0.464
Median R^2	0.338	0.357	0.385	0.358	0.403	0.453	0.465	0.482
Obs.	580208	580208	580208	580208	580208	580208	580208	580208
Bonds	8327	8327	8327	8327	8327	8327	8327	8327
Panel B: Principal Component Analysis								
FVE		0.160	0.171	0.075	0.315	0.250	0.239	0.233
PC1	0.797	0.755	0.743	0.784	0.703	0.700	0.689	0.667
PC2	0.065	0.072	0.077	0.071	0.088	0.091	0.094	0.101
UV	0.752	0.632	0.623	0.696	0.515	0.427	0.393	0.337
Panel C: Time-Series Regression of PC1 on Inflation-Risk Proxies								
Adj. R^2		0.115	0.176	0.050	0.223	0.152	0.130	0.095
R^2		0.118	0.180	0.054	0.232	0.162	0.141	0.106
F-stat		32.424	52.985	13.744	24.219	15.492	13.144	9.525
p-value		0.000	0.000	0.000	0.000	0.000	0.000	0.000
Obs		244	244	244	244	244	244	244
Panel D: Fixed Effects Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.434^{***}			-0.352^{***}	-0.341^{***}	-0.308^{***}	-0.291^{***}
		(-3.444)			(-4.083)	(-3.835)	(-3.816)	(-3.927)
$\Delta\sigma_{i,t}^S$			1.524^{***}		1.151^{***}	1.021^{***}	0.944^{***}	0.723^{***}
			(3.521)		(3.917)	(3.989)	(3.685)	(3.255)
$\Delta\rho_t^S$				0.606^{**}	0.392^{**}	0.305^{**}	0.304^{***}	0.198^{*}
				(2.351)	(2.493)	(2.443)	(2.646)	(1.949)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Adj. R^2	0.145	0.161	0.163	0.151	0.176	0.179	0.185	0.192
R^2	0.157	0.173	0.175	0.163	0.188	0.191	0.197	0.203

REFERENCES

Andreasen, M.M., Christensen, J.H.E., Riddell, S., 2021. The TIPS liquidity premium. Review of Finance 25, 1639–1675. URL: https://academic.oup.com/rof/article/25/6/1639/6410663, doi:https://doi.org/10.1093/rof/rfab018.

Bahaj, S., Czech, R., Ding, S., Reis, R., 2023. The market for inflation risk. Staff Working Paper, 1028, Bank of England URL: https://www.bankofengland.co.uk/working-paper/2023/the-market-for-inflation-risk.

Baldwin, C.Y., Ruback, R.S., 1986. Inflation, uncertainty, and investment. The Journal of Finance 41, 657–668. URL: https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6261.1986.tb04528.x, doi:https://doi.org/10.2307/2328496.

Bessembinder, H., Kahle, K.M., Maxwell, W.F., Xu, D., 2008. Measuring abnormal bond performance. Review of Financial Studies 22, 4219–4258. URL: http://dx.doi.org/10.1093/rfs/hhn105, doi:10.1093/rfs/hhn105.

Bhamra, H.S., Dorion, C., Jeanneret, A., Weber, M., 2022. High inflation: Low default risk and low equity valuations. The Review of Financial Studies 36, 1192–1252. URL: http://dx.doi.org/10.1093/rfs/hhac021, doi:10.1093/rfs/hhac021.

Bhamra, H.S., Fisher, A.J., Kuehn, L.A., 2011. Monetary policy and corporate default. Journal of Monetary Economics 58, 480–494. URL: http://dx.doi.org/10.1016/j.jmoneco.2011.05.010, doi:10.1016/j.jmoneco.2011.05.010.

Bonelli, D., Palazzo, B., Yamarthy, R., 2025. Good inflation, bad inflation, and the dynamics of credit risk. Working Paper URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4798269.

Boons, M., Duarte, F., de Roon, F., Szymanowska, M., 2020. Time-varying inflation risk and stock returns. Journal of Financial Economics 136, 444–470. URL: https://www.

sciencedirect.com/science/article/abs/pii/S0304405X19302429, doi:10.1016/j.jfineco.2019.10.004.

Bu, C., Rogers, J., Wu, W., 2021. A unified measure of Fed monetary policy shocks. Journal of Monetary Economics 118, 331–349. URL: http://dx.doi.org/10.1016/j.jmoneco.2020.11.002, doi:10.1016/j.jmoneco.2020.11.002.

Campbell, J.Y., Pflueger, C., Viceira, L.M., 2020. Macroeconomic drivers of bond and equity risks. Journal of Political Economy 128, 3148–3185. URL: http://dx.doi.org/10.1086/707766, doi:10.1086/707766.

Campbell, J.Y., Shiller, R.J., Viceira, L.M., 2012. Understanding inflation-indexed bond markets , 29–70URL: http://dx.doi.org/10.7551/mitpress/9037.003.0005, doi:10.7551/mitpress/9037.003.0005.

Campbell, J.Y., Taksler, G.B., 2003. Equity volatility and corporate bond yields. Journal of Finance 58, 2321–2349. doi:10.1046/j.1540-6261.2003.00607.x.

Ceballos, L., 2021. Inflation volatility risk and the cross-section of corporate bond returns. Working Paper URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3883556.

Chen, H., 2010. Macroeconomic conditions and the puzzles of credit spreads and capital structure. The Journal of Finance 65, 2171–2212. URL: http://dx.doi.org/10.1111/j.1540-6261.2010.01613.x, doi:10.1111/j.1540-6261.2010.01613.x.

Chen, N.F., Roll, R., Ross, S.A., 1986. Economic forces and the stock market. The Journal of Business 59, 383. URL: http://dx.doi.org/10.1086/296344, doi:10.1086/296344.

Christensen, J.H.E., Lopez, J.A., Rudebusch, G.D., 2015. Pricing deflation risk with US treasury yields. Review of Finance 20, 1107–1152. URL: http://dx.doi.org/10.1093/rof/rfv029, doi:10.1093/rof/rfv029.

Collin-Dufresne, P., Goldstein, R.S., Martin, J.S., 2001. The determinants of credit spread changes. The Journal of Finance 56, 2177–2207. URL: http://dx.doi.org/10.1111/0022-1082.00402, doi:10.1111/0022-1082.00402.

David, A., 2007. Inflation uncertainty, asset valuations, and the credit spreads puzzle. Review of Financial Studies 21, 2487–2534. URL: http://dx.doi.org/10.1093/rfs/hhm041, doi:10.1093/rfs/hhm041.

David, A., Veronesi, P., 2013. What ties return volatilities to price valuations and fundamentals? Journal of Political Economy 121, 682–746. URL: http://dx.doi.org/10.1086/671799, doi:10.1086/671799.

Diamond, D.W., 1991. Debt maturity structure and liquidity risk. Quarterly Journal of Economics 106, 709–737. doi:10.2307/2937924.

Dick-Nielsen, J., Poulsen, T.K., 2019. How to clean academic TRACE data. Working Paper URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3456082.

Diercks, A.M., Campbell, C., Sharpe, S.A., Soques, D., 2023. The swaps strike back: Evaluating expectations of one-year inflation. Finance and Economics Discussion Series , 1–43URL: http://dx.doi.org/10.17016/s.2023.061, doi:10.17016/s.2023.061.

Dovern, J., Fritsche, U., Slacalek, J., 2012. Disagreement among forecasters in g7 countries. The Review of Economics and Statistics 94, 1081–1096. URL: https://EconPapers.repec.org/RePEc:tpr:restat:v:94:y:2012:i:4:p:1081-1096.

Du, D., Elkamhi, R., Ericsson, J., 2019. Time-varying asset volatility and the credit spread puzzle. The Journal of Finance 74, 1841–1885. URL: http://dx.doi.org/10.1111/jofi.12765, doi:10.1111/jofi.12765.

D'Amico, S., Kim, D.H., Wei, M., 2018. Tips from TIPS: The informational content of treasury inflation-protected security prices. Journal of Financial and Quantitative Analysis 53, 395–436. URL: http://dx.doi.org/10.1017/s0022109017000916, doi:10.1017/s0022109017000916.

Edwards, A.K., Harris, L.E., Piwowar, M.S., 2007. Corporate bond market transaction costs and transparency. The Journal of Finance 62, 1421–1451. URL: http://dx.doi.org/10.1111/j.1540-6261.2007.01240.x, doi:10.1111/j.1540-6261.2007.01240.x.

Eisfeldt, A.L., Herskovic, B., Liu, S., 2024. Interdealer price dispersion and intermediary capacity. NBER Working Paper, 32998, National Bureau of Economic Research URL: https://www.nber.org/system/files/working_papers/w32998/w32998.pdf.

Eraker, B., Shaliastovich, I., Wang, W., 2015. Durable goods, inflation risk, and equilibrium asset prices. The Review of Financial Studies 29, 193–231. URL: https://doi.org/10.1093/rfs/hhv049, doi:10.1093/rfs/hhv049, arXiv:https://academic.oup.com/rfs/article-pdf/29/1/193/24450725/hhv049.pdf.

Fama, E., 1981. Stock returns, real activity, inflation, and money. American Economic Review 71, 545–565. URL: https://www.jstor.org/stable/1806180.

Fang, X., Liu, Y., Roussanov, N., 2025. Getting to the core: Inflation risks within and across asset classes. The Review of Financial Studies URL: https://doi.org/10.1093/rfs/hhaf050, doi:10.1093/rfs/hhaf050.

Fischer, G., 2016. Investment choice and inflation uncertainty. London School of Economics and Political Science, Working Paper URL: http://eprints.lse.ac.uk/54259/.

Fleckenstein, M., Longstaff, F., Lustig, H., 2016. Inflation-adjusted bonds and the inflation risk premium , 41–52URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118709207.ch3, doi:https://doi.org/10.1002/9781118709207.ch3, arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781118709207.ch3.

Fleckenstein, M., Longstaff, F.A., Lustig, H., 2014. The TIPS-treasury bond puzzle. The Journal of Finance 69, 2151–2197. URL: http://dx.doi.org/10.1111/jofi.12032, doi:10.1111/jofi.12032.

Fleckenstein, M., Longstaff, F.A., Lustig, H., 2017. Deflation risk. The Review of Financial Studies 30, 2719–2760. URL: http://dx.doi.org/10.1093/rfs/hhx021, doi:10.1093/rfs/hhx021.

Fleming, M.J., Sporn, J., 2013. Trading activity and price transparency in the inflation swap market. Economic Policy Review 19, 1–20. URL: https://www.newyorkfed.org/medialibrary/media/research/epr/2013/0513flem.pdf.

Friewald, N., Nagler, F., 2019. Over-the-counter market frictions and yield spread changes. The Journal of Finance 74, 3217–3257. URL: http://dx.doi.org/10.1111/jofi.12827, doi:10.1111/jofi.12827.

Friewald, N., Nagler, F., Wagner, C., 2022. Debt refinancing and equity returns. The Journal of Finance 77, 2287–2329. URL: http://dx.doi.org/10.1111/jofi.13162, doi:10.1111/jofi.13162.

Goldstein, R., Ju, N., Leland, H., 2001. An ebit-based model of dynamic capital structure. Journal of Business 74, 483–512. doi:10.1086/209679.

Gomes, J.a., Jermann, U., Schmid, L., 2016. Sticky leverage. American Economic Review 106, 3800–3828. URL: http://dx.doi.org/10.1257/aer.20130952, doi:10.1257/aer.20130952.

Gomes, J.a., Schmid, L., 2020. Equilibrium asset pricing with leverage and default. The Journal of Finance 76, 977–1018. URL: http://dx.doi.org/10.1111/jofi.12987, doi:10.1111/jofi.12987.

Grishchenko, O.V., Vanden, J.M., Zhang, J., 2016. The informational content of the embedded deflation option in TIPS. Journal of Banking & Finance 65, 1–26. URL: http://dx.doi.org/10.1016/j.jbankfin.2015.12.004, doi:10.1016/j.jbankfin.2015.12.004.

Gürkaynak, R.S., Sack, B., Wright, J.H., 2007. The U.S. treasury yield curve: 1961 to the present. Journal of Monetary Economics 54, 2291–2304. URL: http://dx.doi.org/10.1016/j.jmoneco.2007.06.029, doi:10.1016/j.jmoneco.2007.06.029.

Gürkaynak, R.S., Sack, B., Wright, J.H., 2010. The TIPS yield curve and inflation compensation. American Economic Journal: Macroeconomics 2, 70–92. URL: http://dx.doi.org/10.1257/mac.2.1.70, doi:10.1257/mac.2.1.70.

Haubrich, J., Pennacchi, G., Ritchken, P., 2012. Inflation expectations, real rates, and risk premia: Evidence from inflation swaps. Review of Financial Studies 25, 1588–1629. URL: http://dx.doi.org/10.1093/rfs/hhs003, doi:10.1093/rfs/hhs003.

He, Z., Khorrami, P., Song, Z., 2022. Commonality in credit spread changes: Dealer inventory and intermediary distress. The Review of Financial Studies 35, 4630–4673. URL: http://dx.doi.org/10.1093/rfs/hhac004, doi:10.1093/rfs/hhac004.

Kang, J., Pflueger, C.E., 2015. Inflation risk in corporate bonds. The Journal of Finance 70, 115–162. URL: https://onlinelibrary.wiley.com/doi/10.1111/jofi.12187, doi:10.1111/jofi.12187.

Leland, H.E., 1998. Agency costs, risk management, and capital structure. The Journal of Finance 53, 1213–1243. URL: http://dx.doi.org/10.1111/0022-1082.00051, doi:10.1111/0022-1082.00051.

Lu, X., Nozawa, Y., Song, Z., 2025. Inflation, default, and corporate bond returns. Johns Hopkins Carey Business School Research Paper URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5349911, doi:10.2139/ssrn.5349911.

Nelson, C.R., Siegel, A.F., 1987. Parsimonious modeling of yield curves. Journal of Business 60, 473–489. doi:10.1086/296409.

Reis, R., 2020. The People versus the Markets: A Parsimonious Model of Inflation Expectations. Discussion Papers 2033. Centre for Macroeconomics (CFM). URL: https://ideas.repec.org/p/cfm/wpaper/2033.html.

Sack, B., Elsasser, R., 2002. Treasury inflation-indexed debt: A review of the U.S. experience. Finance and Economics Discussion Series 2002.0, 1–40. URL: http://dx.doi.org/10.17016/feds.2002.32, doi:10.17016/feds.2002.32.

Svensson, L.E.O., 1994. Estimating and interpreting forward interest rates: Sweden 1992-1994. URL: https://www.elibrary.imf.org/view/journals/001/1994/114/001.1994.issue-114-en.xml, doi:10.5089/9781451853759.001.

Weber, M., 2014. Nominal rigidities and asset pricing. Working Paper URL: https://red-files-public.s3.amazonaws.com/meetpapers/2014/paper_53.pdf.

Appendix

In this appendix, I introduce a structural model of default with debt rollover, stochastic inflation, sticky leverage, and incomplete nominal pass-through, and study its implications for yield spreads. Firms issue nominal debt while cash flows are driven by real activity and inflation. The mismatch between nominal liabilities and imperfectly adjusting assets links inflation risk to default risk. The model rationalizes the empirical findings that yield spreads decrease with expected inflation, increase with inflation uncertainty, and depend on the correlation between inflation and real asset innovations. It also generates heterogeneity: sensitivity to inflation risk increases with default risk and the degree of nominal pass-through, denoted by ϕ, and decreases with refinancing intensity.

Inflation Risk. Let P_t denote the price level and A_t^r real firm value. Nominal firm value is given by

$$A_t^n = A_t^r P_t^{\phi}, \tag{2}$$

where $\phi \in [0,1]$ captures the degree of pass-through of inflation into firm value. When $\phi < 1$, assets are sticky and adjust only partially to changes in the price level.

Inflation evolves under the physical measure $\mathbb{P}$ as

$$\frac{dP_t}{P_t} = \mu_P^{\mathbb{P}} dt + \sigma_P dW_t^{P,\mathbb{P}}, \tag{3}$$

where $\mu_P^{\mathbb{P}}$ denotes physical expected inflation and σ_P denotes inflation volatility. I allow for a constant inflation risk premium through the market price of inflation risk λ_P. The change of measure from $\mathbb{P}$ to the nominal risk-neutral measure $\mathbb{Q}^n$ is

$$dW_t^{P,\mathbb{Q}^n} = dW_t^{P,\mathbb{P}} + \lambda_P dt. \tag{4}$$

Substituting this into the physical inflation process gives the risk-neutral inflation process

$$\frac{dP_t}{P_t} = \mu_p dt + \sigma_P dW_t^{P,\mathbb{Q}^n}, \tag{5}$$

where

$$\mu_p = \mu_P^{\mathbb{P}} - \sigma_P \lambda_P. \tag{6}$$

Thus, μ_p is the risk-neutral expected inflation rate. It differs from physical expected inflation because investors require compensation for bearing inflation risk.

With a constant real risk-free rate r_r, the nominal risk-free rate is

$$r_n = r_r + \mu_p = r_r + \mu_P^{\mathbb{P}} - \sigma_P \lambda_P. \tag{7}$$

Real asset growth and inflation shocks are correlated under the nominal risk-neutral measure:

$$dW_t^{A^r,\mathbb{Q}^n} dW_t^{P,\mathbb{Q}^n} = \rho_{A^r P} dt. \tag{8}$$

Applying Itô's lemma to $A_t^n = A_t^r P_t^{\phi}$, nominal asset volatility is

$$\sigma_{A^n}^2 = \sigma_{A^r}^2 + \phi^2 \sigma_P^2 + 2\phi \rho_{A^r P} \sigma_{A^r} \sigma_P. \tag{9}$$

Under the nominal risk-neutral measure, nominal firm value follows

$$\frac{dA_t^n}{A_t^n} = r_n dt + \sigma_{A^n} dW_t^{A^n, \mathbb{Q}^n}. \tag{10}$$

I set the asset payout rate equal to zero in the baseline model. If assets pay out at rate δ_A, the drift becomes $r_n - \delta_A$.

Expected inflation affects pricing through the risk-neutral expected inflation rate μ_p and therefore through the nominal risk-free rate r_n. The inflation risk premium affects pricing through the wedge between $\mu_P^{\mathbb{P}}$ and μ_p. Inflation volatility affects firm value through σ_{A^n}, with exposure governed by ϕ and $\rho_{A^r P}$.

Debt Structure and Yield Spreads. The firm issues perpetual debt with coupon C and continuously refinances a fraction m of face value P (Leland, 1998). Default occurs when nominal assets fall below a threshold:

$$\tau = \inf\{t \geq 0 : A_t^n \leq A_B^n\}. \tag{11}$$

The value of a unit claim paid at default is

$$P_B(A^n) = \left(\frac{A^n}{A_B^n}\right)^{-\gamma}, \tag{12}$$

where

$$\gamma = \frac{r_n - \frac{1}{2}\sigma_{A^n}^2 + \sqrt{\left(r_n - \frac{1}{2}\sigma_{A^n}^2\right)^2 + 2(r_n + m)\sigma_{A^n}^2}}{\sigma_{A^n}^2}. \tag{13}$$

Debt value is

$$D(A^n) = \frac{C + mP}{r_n + m}\left(1 - P_B(A^n)\right) + R A_B^n P_B(A^n). \tag{14}$$

Yield spreads are

$$y - r_n = \frac{C^*}{D(A^n; C^*, P)} - r_n. \tag{15}$$

Model Implications. Yield spreads decrease in expected inflation because higher inflation reduces the real burden of fixed nominal debt. Inflation volatility increases spreads by raising nominal asset risk. This effect depends on $\rho_{A^r P}$. When $\rho_{A^r P} > 0$, inflation shocks and real asset shocks reinforce each other in nominal asset volatility. When $\rho_{A^r P} < 0$, inflation partially hedges real asset risk.

The correlation between inflation and real asset growth also matters. With positive correlation, low real asset growth tends to coincide with low inflation. Since liabilities are fixed in nominal terms, low inflation increases the real burden of debt precisely when firm value is weak. This strengthens the impact of inflation risk on credit spreads.

The model generates cross-sectional variation in inflation exposure. Firms with higher asset volatility are closer to default and exhibit stronger spread sensitivity to inflation risk. Lower refinancing intensity increases exposure to fixed nominal debt and amplifies the effect of expected inflation. Pass-through also matters. When ϕ is high, inflation risk is more strongly transmitted to nominal asset volatility. When ϕ is low, this effect is dampened. Consequently, firms with flexible cash flows are more sensitive to inflation volatility.

Overall, the model implies that inflation risk has a larger impact on firms with higher default probability, lower refinancing intensity, and stronger exposure of nominal asset values to inflation shocks.

Internet Appendix for:
”Inflation Risk and Yield Spread Changes”

Diego Bonelli

This Internet Appendix contains supplemental material for the article ”Inflation Risk and Yield Spread Changes”.

Section *A* presents the main model solution. Section *B* reports a description of the inflation swap market and *C* details the procedure for seasonal adjustment of swap and TIPS rates. Section *D* details the construction of the TRACE to CRSP linking table. Section *E* presents additional tables.

- Table I presents model parameters.
- Tables II and III report replication tables of CDGM.
- Tables IV, V, and VI report heterogeneity by firm fundamentals (size, ROA, profit margin, rating), operating channels (operating leverage, markup, input-cost exposure), and debt structure (leverage, refinancing intensity, debt growth, floating-rate, maturity), respectively.
- Table VII presents time-series regressions including all bonds regardless of industry, size, or type of bond.
- Table VIII presents time-series regressions including only bonds which trade during the last trading day of the month as in Friewald and Nagler (2019).

A. Model Solution

Let A_t^r and P_t denote the firm's real asset value and the price index, respectively. I define nominal asset value as

$$A_t^n = A_t^r P_t^{\phi}, \tag{IA.1}$$

where $0 < \phi \leq 1$ captures the sensitivity of nominal assets to inflation. When $\phi = 1$, nominal assets fully adjust to inflation, whereas when $\phi < 1$, pass-through is incomplete and assets are sticky in nominal terms.

I assume that inflation follows a geometric Brownian motion under the physical measure $\mathbb{P}$:

$$\frac{dP_t}{P_t} = \mu_P^{\mathbb{P}} dt + \sigma_P dW_t^{P,\mathbb{P}}. \tag{IA.2}$$

The market price of inflation risk is denoted by λ_P. The change of measure from $\mathbb{P}$ to the nominal risk-neutral measure $\mathbb{Q}^n$ is

$$dW_t^{P,\mathbb{Q}^n} = dW_t^{P,\mathbb{P}} + \lambda_P dt. \tag{IA.3}$$

Therefore, under $\mathbb{Q}^n$,

$$\frac{dP_t}{P_t} = \mu_p dt + \sigma_P dW_t^{P,\mathbb{Q}^n}, \tag{IA.4}$$

where

$$\mu_p = \mu_P^{\mathbb{P}} - \sigma_P \lambda_P. \tag{IA.5}$$

The variable μ_p is the risk-neutral expected inflation rate. Since the debt claims in the model are nominal, the relevant pricing measure is the nominal risk-neutral measure $\mathbb{Q}^n$.

Let r_r denote the real risk-free rate. The nominal risk-free rate is

$$r_n = r_r + \mu_p. \tag{IA.6}$$

Under the nominal risk-neutral measure, I write the real asset process as

$$\frac{dA_t^r}{A_t^r} = \delta_{A^r}^n dt + \sigma_{A^r} dW_t^{A^r,\mathbb{Q}^n}, \tag{IA.7}$$

where the Brownian motions $dW_t^{A^r,\mathbb{Q}^n}$ and $dW_t^{P,\mathbb{Q}^n}$ have instantaneous correlation

$$dW_t^{A^r,\mathbb{Q}^n} dW_t^{P,\mathbb{Q}^n} = \rho_{A^r P} dt. \tag{IA.8}$$

Since

$$A_t^n = A_t^r P_t^{\phi}, \tag{IA.9}$$

Itô's lemma implies

$$\frac{dA_t^n}{A_t^n} = \frac{dA_t^r}{A_t^r} + \phi \frac{dP_t}{P_t} + \phi \frac{dA_t^r}{A_t^r} \frac{dP_t}{P_t} + \frac{1}{2} \phi(\phi - 1) \sigma_P^2 dt. \tag{IA.10}$$

Substituting the processes for A_t^r and P_t gives

$$\frac{dA_t^n}{A_t^n} = \left[\delta_{A^r}^n + \phi\mu_p + \phi\rho_{A^rP}\sigma_P\sigma_{A^r} + \frac{1}{2}\phi(\phi-1)\sigma_P^2\right]dt + \sigma_{A^r}dW_t^{A^r,\mathbb{Q}^n} + \phi\sigma_P dW_t^{P,\mathbb{Q}^n}. \tag{IA.11}$$

Because A_t^n represents the value of traded assets in nominal units, its expected return under $\mathbb{Q}^n$ must equal the nominal risk-free rate. In the baseline model, I set the asset payout rate equal to zero. Therefore, the nominal asset process satisfies

$$\frac{dA_t^n}{A_t^n} = r_n dt + \sigma_{A^n}dW_t^{A^n,\mathbb{Q}^n}. \tag{IA.12}$$

Matching the drift of the Itô expression to $r_n = r_r + \mu_p$ implies

$$\delta_{A^r}^n = r_r + (1-\phi)\mu_p - \phi\rho_{A^rP}\sigma_P\sigma_{A^r} - \frac{1}{2}\phi(\phi-1)\sigma_P^2. \tag{IA.13}$$

Thus, under the nominal risk-neutral measure, the real asset process follows

$$\frac{dA_t^r}{A_t^r} = \left[r_r + (1-\phi)\mu_p - \phi\rho_{A^rP}\sigma_P\sigma_{A^r} - \frac{1}{2}\phi(\phi-1)\sigma_P^2\right]dt + \sigma_{A^r}dW_t^{A^r,\mathbb{Q}^n}. \tag{IA.14}$$

Substituting this expression back into the nominal asset dynamics gives

$$\frac{dA_t^n}{A_t^n} = r_n dt + \sigma_{A^r}dW_t^{A^r,\mathbb{Q}^n} + \phi\sigma_P dW_t^{P,\mathbb{Q}^n}. \tag{IA.15}$$

Defining total nominal asset variance as

$$\sigma_{A^n}^2 = \sigma_{A^r}^2 + \phi^2\sigma_P^2 + 2\phi\rho_{A^rP}\sigma_{A^r}\sigma_P, \tag{IA.16}$$

I can write the nominal asset process compactly as

$$\frac{dA_t^n}{A_t^n} = r_n dt + \sigma_{A^n}dW_t^{A^n,\mathbb{Q}^n}. \tag{IA.17}$$

The firm commits to a stationary debt structure by issuing consol bonds to optimally set total debt through a constant coupon rate C. Following Leland (1998), the firm continuously retires a fraction mP of its outstanding debt at par, where P denotes the total face value of debt and m is the refinancing intensity, with $0 < m \leq 1$. The retired debt is immediately replaced with newly issued debt of identical maturity, coupon, principal, and seniority, which ensures a constant rollover structure.

Debt issuance provides a tax advantage $\tau_{tax}C$ but also increases bankruptcy costs borne by equity holders. Coupon payments can be funded either out of cash flow plus the tax shield or, if necessary, out of equity holders' own resources. Default occurs when equity holders deem firm value too low to justify further payments. Upon liquidation, debt holders recover the residual value of the firm net of bankruptcy costs, determined by the recovery rate R.

Equity holders are the residual claimants on the firm's cash flows. Therefore, equity value,

E, is given by the difference between levered firm value, ν, and debt value, D. If A_t^n declines to a critical endogenous threshold, A_B^n, equity holders default. Let τ denote the first time assets hit A_B^n:

$$\tau = \inf\{t \geq 0 : A_t^n \leq A_B^n\}. \tag{IA.18}$$

Let $r_n = r_r + \mu_p$. The value of a unit claim at default is

$$\begin{aligned} P_B(A^n) &= \left(\frac{A^n}{A_B^n}\right)^{-\gamma}, \\ \gamma &= \frac{r_n - \frac{1}{2}\sigma_{A^n}^2 + \sqrt{\left(r_n - \frac{1}{2}\sigma_{A^n}^2\right)^2 + 2(r_n+m)\sigma_{A^n}^2}}{\sigma_{A^n}^2}. \end{aligned} \tag{IA.19}$$

Debt value is then given by

$$D(A^n; A_B^n, P, C) = \frac{C + mP}{r_n + m}\left[1 - P_B(A^n)\right] + RA_B^n P_B(A^n), \tag{IA.20}$$

where the first term is the value of coupon and principal payments up to default and the second term is the recovery value in bankruptcy.

The total value of the firm, ν, equals asset value plus tax benefits minus bankruptcy costs:

$$\nu(A^n; A_B^n, P, C) = A^n + TB(A^n; A_B^n, C) - BC(A^n; A_B^n). \tag{IA.21}$$

Define $x = \gamma$ when $m = 0$. Equivalently,

$$x = \frac{r_n - \frac{1}{2}\sigma_{A^n}^2 + \sqrt{\left(r_n - \frac{1}{2}\sigma_{A^n}^2\right)^2 + 2r_n\sigma_{A^n}^2}}{\sigma_{A^n}^2}. \tag{IA.22}$$

Then the present value of bankruptcy costs is

$$BC(A^n; A_B^n) = (1-R)A_B^n \left(\frac{A^n}{A_B^n}\right)^{-x}. \tag{IA.23}$$

The value of the coupon stream up to default is

$$\frac{C}{r_n}\left[1 - \left(\frac{A^n}{A_B^n}\right)^{-x}\right], \tag{IA.24}$$

which implies a tax benefit of debt equal to

$$TB(A^n; A_B^n, C) = \frac{\tau_{tax}C}{r_n}\left[1 - \left(\frac{A^n}{A_B^n}\right)^{-x}\right]. \tag{IA.25}$$

Since the firm has only equity and debt outstanding, firm value satisfies

$$\begin{aligned} \nu(A^n;A_B^n,P,C) &= E(A^n;A_B^n,P,C)+D(A^n;A_B^n,P,C) \\ &= A^n+TB(A^n;A_B^n,C)-BC(A^n;A_B^n). \end{aligned} \tag{IA.26}$$

Since $E=\nu-D$, equity value is

$$\begin{aligned} E(A^n;A_B^n,P,C) =& A^n+\frac{\tau_{tax}C}{r_n}\left[1-\left(\frac{A^n}{A_B^n}\right)^{-x}\right]-(1-R)A_B^n\left(\frac{A^n}{A_B^n}\right)^{-x} \\ &-\frac{C+mP}{r_n+m}\left[1-P_B(A^n)\right]-RA_B^nP_B(A^n). \end{aligned} \tag{IA.27}$$

The initial owners of the assets choose the debt level that maximizes total firm value. Since equity is a limited-liability claim, equity holders never allow equity value to become negative. Instead, they stop servicing debt and force the firm into bankruptcy. Therefore, the optimal default-triggering level satisfies the smooth-pasting condition

$$\frac{d}{dA^n}E(A^n;A_B^n,P,C)\bigg|_{A^n=A_B^n}=0. \tag{IA.28}$$

This condition yields

$$A_B^n=\frac{\frac{\gamma(C+mP)}{r_n+m}-\frac{\tau_{tax}Cx}{r_n}}{1+(1-R)x+R\gamma}. \tag{IA.29}$$

Plugging this solution into equation (IA.20) gives a closed-form expression for total debt value, conditional on coupon C and principal value P. When debt is initially issued, there is an additional restriction linking market value, coupon, and principal: the coupon is set such that market debt value, D, equals principal value, P. That is, debt is issued at par. This requires C to solve

$$D(A^n;A_B^n,P,C)=P. \tag{IA.30}$$

At time $t=0$, the initial owners choose the optimal debt policy to maximize total firm value. Equivalently, for each candidate principal P, the coupon is chosen to satisfy the par-pricing condition, and the firm selects the debt policy that maximizes firm value:

$$(C^*,P^*)=\arg\max_{C,P}\nu(A_0^n;A_B^n(C,P),P,C), \tag{IA.31}$$

subject to

$$D(A_0^n;A_B^n(C,P),P,C)=P. \tag{IA.32}$$

Yield spreads are therefore given by

$$y-r_n=\frac{C^*}{D(A_0^n;A_B^n(C^*,P^*),P^*,C^*)}-r_n. \tag{IA.33}$$

When debt is issued at par, this simplifies to

$$y - r_n = \frac{C^*}{P^*} - r_n. \tag{IA.34}$$

A.1. Model Implications

To illustrate how yield spreads relate to inflation risk, I simulate the model 1000 times by drawing parameter values for expected inflation, μ_p, inflation volatility, σ_P, and the correlation between inflation and asset growth, ρ_{A^rP}, from reasonable parameter intervals. Throughout the simulations, I set the inflation risk premium to zero, $\lambda_P = 0$, so physical and risk-neutral expected inflation coincide. Expected inflation ranges from -2% to 10%, inflation volatility from 0.5% to 10%, and the correlation between inflation and assets from -1 to 1. All other parameters are fixed at values commonly used in the literature, as summarized in Table I.[i]

Table I about here

A.1.1. Implications for Yield Spreads

Figure 1 illustrates the relationship between inflation risk and yield spreads, together with the corresponding regression lines. Blue dots denote observations for which the inflation-asset growth correlation, ρ_{A^rP}, is positive, while red dots denote observations for which it is negative.

Figure 1 about here

Panel A shows that yield spreads decrease with expected inflation. An increase in expected inflation lowers the real value of nominal debt, since debt is denominated in nominal terms. This reduction in the real burden of debt lowers default risk and, in turn, narrows yield spreads.

Panel B shows the effect of inflation volatility on yield spreads. Higher inflation volatility increases the volatility of nominal assets, thereby raising default risk and widening spreads. The strength of this effect depends on the correlation between inflation and real asset growth. When asset-inflation correlation is positive, the direct effect of inflation volatility and the covariance term in nominal asset volatility reinforce each other, generating a stronger positive relationship between inflation volatility and yield spreads.[ii] When correlation is negative, the covariance term offsets part of the direct volatility effect, making the relationship between yield spreads and inflation volatility flatter. The dispersion in Figure 1, Panel B, reflects this asymmetric response of nominal asset volatility to inflation volatility depending on the sign of the asset-inflation correlation.

Panel C highlights the role of the correlation between inflation and real asset innovations, which captures the cyclicality of inflation risk. When this correlation is positive, inflation shocks and real asset shocks reinforce each other in nominal asset volatility. This increases

[i] The stickiness parameter ϕ is set to 0.4 as in Bhamra, Dorion, Jeanneret and Weber (2022), while the recovery rate R and tax rate τ_{tax} are set to 0.5 and 0.35, respectively, as in Leland (1998) and Du, Elkamhi and Ericsson (2019).

[ii] Since inflation-growth correlation is unconditionally positive in the sample, the amplification mechanism under positive ρ_{A^rP} is the empirically most relevant case.

default risk and widens yield spreads. When the correlation is negative, inflation shocks partially hedge real asset risk, dampening the effect of inflation volatility on spreads.

A.1.2. Heterogeneity Implications

The model also generates predictions about heterogeneity, particularly with respect to asset volatility, the degree of nominal pass-through, ϕ, and refinancing intensity. More generally, it suggests that the effect of inflation risk on a firm depends on both its distance to default and its exposure to inflation shocks.

Figure 2 shows that, holding nominal debt constant, higher asset volatility reduces firm value and raises measured leverage. This moves firms closer to their default boundary and amplifies the sensitivity of spreads to asset risk. As a result, firms with higher leverage experience stronger effects from both expected inflation and inflation volatility.

Figure 2 about here

In addition, as in the model of Friewald, Nagler and Wagner (2022), leverage and refinancing intensity have opposite implications for default probabilities, with lower refinancing intensity being associated with higher default risk. Thus, for a given leverage level, lower refinancing intensity increases the sensitivity of spreads to inflation risk, as shown in Figure 3.

Figure 3 about here

Figure 4 shows that cash-flow or asset stickiness determines the extent to which inflation risk passes through to nominal asset volatility. As shown in the expression for σ_{A^n}, when assets are more flexible, so that ϕ is closer to one, inflation shocks have a larger effect on nominal asset risk. When assets are stickier, so that $\phi < 1$, this effect is dampened. Consequently, firms with more flexible cash flows are more sensitive to inflation volatility and to the interaction between inflation and real asset growth than firms with stickier cash flows.

Figure 4 about here

The empirical variables in Section III are not direct measures of the model's pass-through parameter ϕ. The model uses ϕ as a reduced-form parameter governing how inflation shocks enter nominal asset values. In the data, markup and input-cost exposure capture related but distinct margins of operating exposure: markup captures the room firms have to absorb cost shocks without compressing margins, while input-cost exposure captures how strongly firms' cash flows are tied to inflationary production costs and sectoral price movements. The markup and input-cost exposure patterns documented in Section IV should therefore be interpreted as evidence that operating exposure shapes inflation sensitivity, rather than as a literal test of monotonic variation in ϕ.

Although stylized, the model captures the key channels through which inflation risk affects yield spreads. In particular, it rationalizes yield spreads decreasing with expected inflation and increasing with inflation volatility and cyclicality. Consistent with the empirical findings, it further implies that inflation risk has a larger impact on firms with higher default probability, lower refinancing intensity, and greater pass-through.

B. Inflation Swaps and Market-Based Inflation Expectations

In 1997, the U.S. Treasury started issuing Treasury Inflation-Protected Securities, fixed coupon bonds whose principal amount is adjusted daily based on the Consumer Price Index (CPI) for All Urban Consumers in the third preceding calendar month. Beginning with the first TIPS auction, market participants began to make markets in inflation derivatives as a way of hedging inflation risk. Zero-coupon inflation swaps quickly became one of the most liquid inflation derivatives in the over-the-counter market. These swaps are forward contracts in which the buyer pays a fixed nominal rate and receives an inflation-linked payment from the seller. They are quoted with maturities ranging from 1 to 30 years and, along with nominal Treasuries, offer an alternative means of measuring real yields.

I use inflation swap rates as a more reliable reflection of market participants' inflation expectations for several reasons. First, inflation swaps tend to predict future inflation rates more accurately than surveys. As shown by Diercks, Campbell, Sharpe and Soques (2023), inflation swaps align more closely with realized inflation rates than survey forecasts, which exhibit less variation. Although inflation swaps display higher variance than survey expectations, they remain less volatile than realized inflation. In Figure 5, consistent with Diercks, Campbell, Sharpe and Soques (2023), I show that surveys tend to cluster most forecasts around 2% compared to the actual realized inflation, while inflation swap rates generally show more variation.

Second, at medium to long horizons, inflation swaps carry only a minimal risk premium component. Bahaj, Czech, Ding and Reis (2023) utilize transaction-level data from UK inflation swaps and show that the supply of long-term inflation protection is highly elastic, reflects economic fundamentals, and rapidly adjusts to new information. Finally, compared to TIPS, inflation swaps offer a more unbiased and reliable measure of inflation expectations due to several key differences. The TIPS inflation adjustment is bounded below at its issuance value providing an embedded put option that protects investors against deflation on the bond's principal payment (e.g., Grishchenko, Vanden and Zhang (2016); Christensen, Lopez and Rudebusch (2015)). Because this option has a nonnegative value, it lowers TIPS yields compared to bonds that are fully indexed to inflation. Zero-coupon inflation swap contracts do not contain this option. Therefore, all else equal, the break-even inflation rate (Treasury rate minus TIPS rate) based on a TIPS principal strip should be higher than the equivalent maturity inflation swap rate. In addition to the deflation option, studies by Sack and Elsasser (2002), Fleckenstein, Longstaff and Lustig (2014), D'Amico, Kim and Wei (2018), and Andreasen, Christensen and Riddell (2021) consistently show that TIPS break-even inflation rates fall below survey-based inflation expectations and that Treasury bonds are almost always overvalued relative to inflation-swapped TIPS. This mispricing narrows as additional capital flows into the markets and as liquidity increases.

Figure 5 about here

Even in the most recent sample, the pattern is consistent with previous studies. Figure 6 reports the inflation swap rate minus TIPS implied break-even rate and shows that it exhibits time variation, with a positive average and peaking during periods of low liquidity. This evidence is consistent with Campbell, Shiller and Viceira (2012) and Haubrich, Pennacchi and Ritchken (2012), who attribute the spike in TIPS yields after Lehman Brothers' bankruptcy to Lehman's extensive use of TIPS for collateralizing its repo borrowings and derivative positions and with Fleckenstein, Longstaff and Lustig (2014) which finds that the price difference narrows when the U.S. auctions nominal Treasuries or TIPS, and it widens when dealers have difficulty obtaining Treasury securities, such as during a period of increased repo failures. Therefore, TIPS yields contain a liquidity premium because, like other bonds, they are held in buy-and-hold investors' portfolios, causing break-even inflation rates to diverge further from inflation swap rates.

Figure 6 about here

Overall, these factors motivate the use of inflation swaps as a more accurate and less biased measure of inflation expectations compared to alternative instruments.

C. Seasonal Adjustment of Swap and TIPS rates

TIPS and zero-coupon inflation swaps are both indexed to the non-seasonally adjusted U.S. CPI index, thus seasonal patterns in inflation must be taken into account when matching with corporate bond cash flows for swap maturities that include fractional years (e.g., 7.5 years). I adjust swap and TIPS rates following the procedure of Fleckenstein, Longstaff and Lustig (2014). Specifically, I fit a standard cubic spline through the quoted maturities of both swaps and TIPS using a grid size of one month. I then estimate seasonal components in inflation from the monthly non-seasonally adjusted U.S. CPI index (CPI-U NSA) series between January 1980 and December 2021 by estimating a regression of monthly log changes in the CPI index on month dummies. I obtain an estimate of the seasonal effect in each month. I normalize these seasonal factors so that their product is unity, thus, by construction, there will be no seasonal adjustment for full-year maturities. Next, monthly forward rates are constructed from the interpolated rates, and multiplied by the corresponding adjustment factor. Lastly, I obtain seasonally adjusted rates by converting forward rates into spot rates. As suggested by Fleckenstein, Longstaff and Lustig (2014), since the interpolated rates would then be sensitive to short-term inflation assumptions, I do not interpolate or adjust maturities shorter than the quoted ones, i.e., one year for the swaps and two years for the TIPS, but instead use the shortest quoted maturity rate.

D. Linking Academic TRACE to CRSP

I create a linking table to match TRACE CUSIPs to CRSP permco, accounting for firm mergers, delistings, and splits. First, I merge TRACE with Mergent FISD to obtain issuer and parent CUSIPs. For the CRSP merge, I rely on 6-digit CUSIPs and the TRACE-reported trading symbol, which corresponds to a firm's ticker.

I begin by matching CRSP to TRACE using issuer CUSIPs, tracking the issuing firm forward through the CRSP delisting table to account for mergers or spin-offs. Trading symbols help resolve cases where a bond corresponds to multiple firms on the same date, prioritizing matches where the TRACE trading symbol aligns with the CRSP company symbol.

For unmatched bonds, I match them to CRSP firms using parent CUSIPs. By tracing the parent firm back in time to the bond's offering date and utilizing trading symbols, I identify the correct firm path. First, I ensure the path reaches the first symbol match before other firms, and if multiple firm-date matches occur, I prioritize matches where the TRACE trading symbol aligns with the CRSP company symbol at the time. If no valid match exists, I use the symbol match at any point, provided no alternative matches exist.

Next, I match firms using trading symbols, requiring that symbols are active simultaneously in both CRSP and TRACE databases. Matches are valid only when dates overlap, and the longest period of valid symbol matches is retained. After removing six instances of multiple firm-bond matches, I match the remaining bonds using company names, first matching the issuing firm from FISD with CRSP at the offering date, and then matching parent firm names for any remaining bonds.

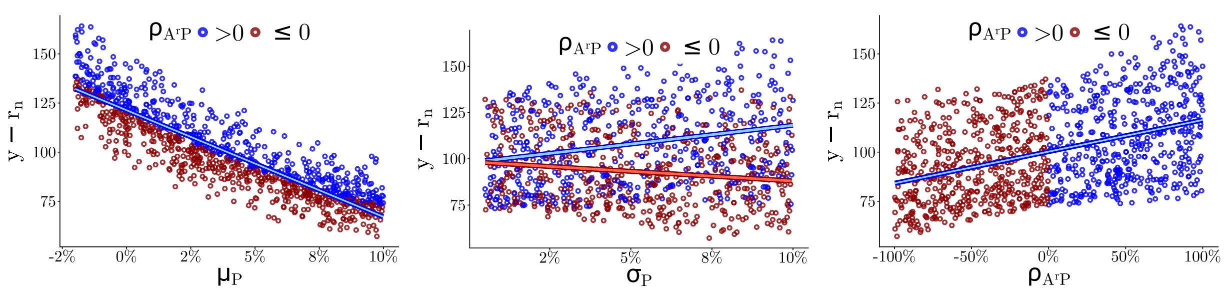
Table I: **Model Parameters**

This table lists the parameters used to examine how yield spreads relate to inflation risk. I fix all parameters except for expected inflation (μ_P), inflation volatility (σ_P) and the correlation between inflation and asset growth ($\rho_{A^r,P}$). I uniformly draw these parameters in reasonable intervals.

Name	Symbol	Value
Initial Price Level	P_0	1
Initial Asset Value	A_0	100
Stickiness Parameter	ϕ	0.40
Refinancing Intensity	m	0.125
Firm-Specific Volatility	σ_{A_r}	0.30
Real Risk-Free Rate	r_r	0.04
Tax Rate	τ_{tax}	0.35
Recovery Rate	R	0.50
Expected Inflation	μ_P	$\mathscr{U}[-0.02, 0.10]$
Correlation Inflation Asset Growth	ρ_{A^rP}	$\mathscr{U}[-1, 1]$
Inflation Volatility	σ_P	$\mathscr{U}[0.005, 0.10]$

Figure 1: **Model Simulation**

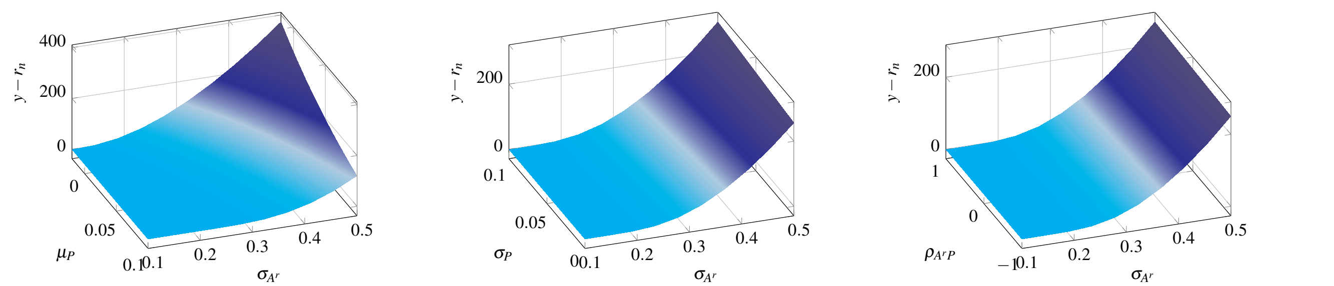
This figure shows the yield spread implications of inflation risk. I simulate the model 1000 times where I fix all parameters except the expected inflation rate (μ_P), inflation volatility (σ_P), and the correlation between inflation and assets (ρ_{A^rP}). I uniformly draw these parameters from reasonable intervals: expected inflation varies between -2% and 10%, inflation volatility between 0.5% and 10%, and correlation between inflation and real assets between -1 and 1. Table I provides the details about all parameter choices. I also report regression lines. In Panel (b), the regression line is conditional on the correlation between inflation and real assets.



(a) Expected Inflation (b) Inflation Volatility (c) Correlation

Figure 2: **Heterogeneity - Asset Volatility**

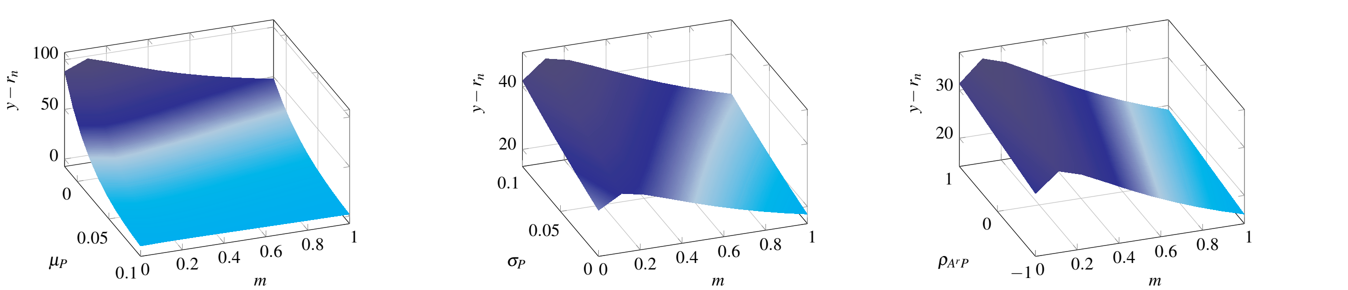
This figure shows the heterogeneity implications of inflation risk. I calibrate the model for each level of real asset volatility from 5 to 50% and for each level of expected inflation rate (μ_P), inflation volatility (σ_P), and the correlation between inflation and assets (ρ_{A^rP}). Expected inflation varies between -2% and 10%, inflation volatility between 0.5% and 10%, and correlation between inflation and real assets between -1 and 1. I fix all parameters according to Table I.



(a) Expected Inflation (b) Inflation Volatility (c) Correlation

Figure 3: **Heterogeneity - Refinancing Intensity**

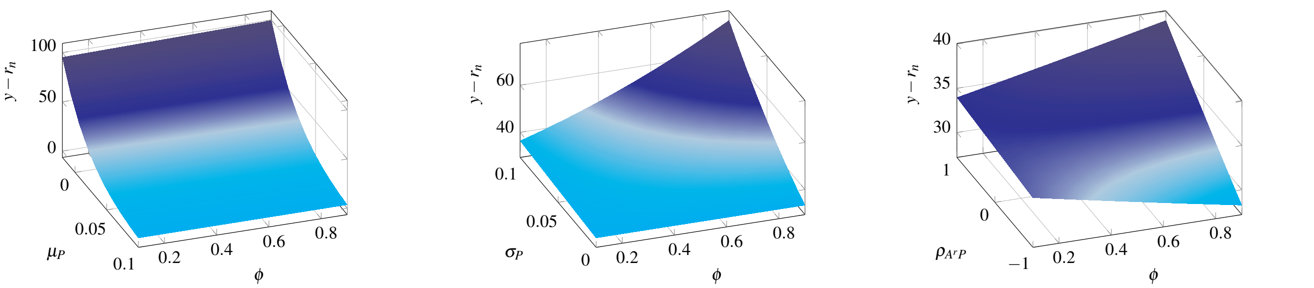
This figure shows the heterogeneity implications of inflation risk. I calibrate the model for each level of refinancing intensity (m) from 0 to 1 and for each level of expected inflation rate (μ_P), inflation volatility (σ_P), and the correlation between inflation and assets (ρ_{A^rP}). Expected inflation varies between -2% and 10%, inflation volatility between 0.5% and 10%, and correlation between inflation and real assets between -1 and 1. I fix all parameters according to Table I.



(a) Expected Inflation (b) Inflation Volatility (c) Correlation

Figure 4: **Heterogeneity - Cash-Flow Stickiness**

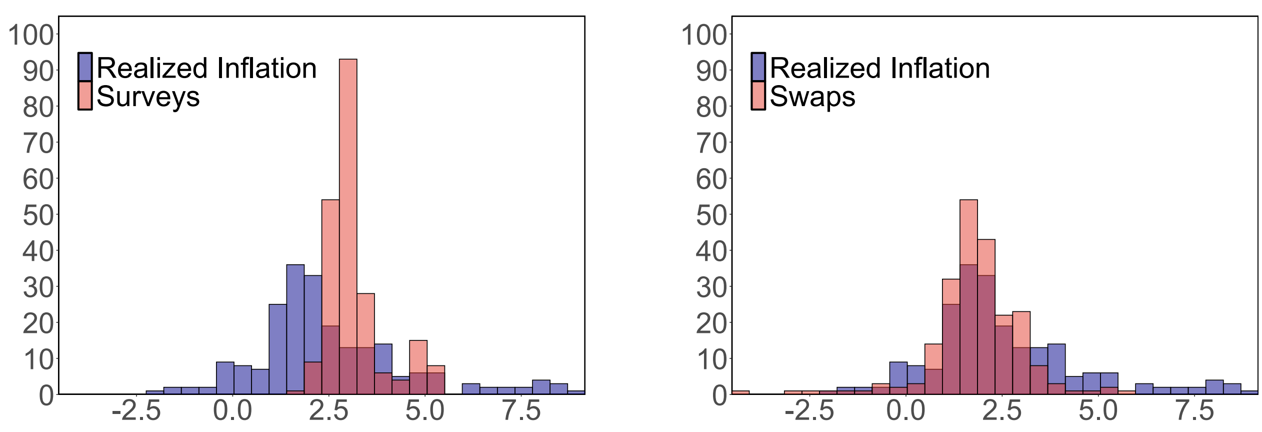
This figure shows the heterogeneity implications of inflation risk. I calibrate the model for each level of the pass-through parameter (ϕ) from 0.1 to 0.9 and for each level of expected inflation rate (μ_P), inflation volatility (σ_P), and the correlation between inflation and assets (ρ_{A^rP}). Expected inflation varies between -2% and 10%, inflation volatility between 0.5% and 10%, and correlation between inflation and real assets between -1 and 1. I fix all parameters according to Table I.



(a) Expected Inflation (b) Inflation Volatility (c) Correlation

Figure 5: **Histogram of Forecasts vs Realized: Surveys and Swaps**

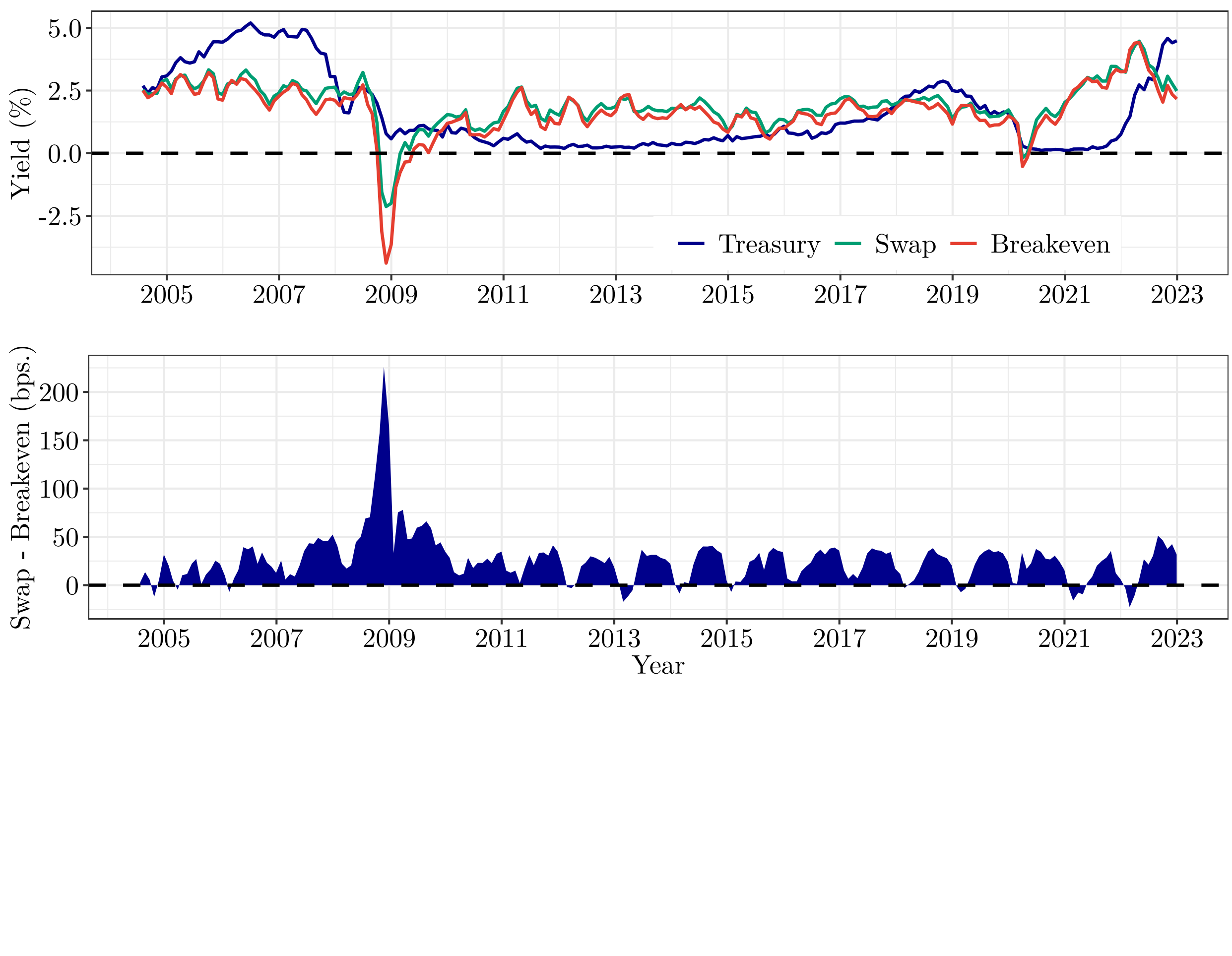
This figure reports the histogram of the forecasts and the 1-year inflation rates. The horizontal axis reflects the forecast or realized inflation in percentage points, while the vertical axis captures the frequency. In blue I report the realizations while in red the expectations. Panel (a) shows the expected price change for one year from Consumer Surveys at the University of Michigan, while Panel (b) shows the one-year inflation swaps.



(a) Surveys (b) Swaps

Figure 6: **Time Series of Treasury, TIPS and Swap rates**

The top panel shows the time series of 2-year zero coupon treasury yield, break-even, and inflation swaps. The bottom panel represents the difference between the 2-year zero coupon inflation swap rate and the 2-year TIPS implied zero-coupon break-even inflation yield. Yields are expressed as annual percentages, and the difference is in annual basis points.



E. Additional Tables

Table II: **Determinants of Yield Spread Changes in Collin-Dufresne, Goldstein and Martin (2001)**

For each industrial bond i with at least 25 monthly observations of yield spread changes, $\Delta YS_{i,t}$, I estimate the model:

$$\Delta YS_{i,t} = \alpha_i + \boldsymbol{\beta}_i^T \boldsymbol{\Delta S}_{i,t} + \varepsilon_{i,t},$$

where $\boldsymbol{\Delta S}_{i,t} := [\Delta Lev_{i,t}, \Delta RF_t, \Delta RF_t^2, \Delta Slope_t, \Delta VIX_t, RM_t, \Delta Jump_t]$ is the vector of the structural model variables, with $\Delta Lev_{i,t}$ as the change in firm leverage, ΔRF_t the change in 10-year Treasury interest rate, ΔRF_t^2 the squared change in the 10-year Treasury interest rate, $\Delta Slope_t$ the change in the slope of the term structure, ΔVIX_t the change in VIX_t index, RM_t the S&P 500 return, and $\Delta Jump_t$ the change in a jump factor based on S&P 500 index options. Panel A reports the average coefficients across bonds, the associated t-statistics, the mean and median adjusted R^2 values, and the numbers of observations and bonds in the sample, respectively. The t-statistics are calculated from the cross-sectional variation over the estimates for each coefficient. That is, each reported coefficient value is divided by the standard deviation of the estimates and scaled by the square root of the number of bonds. The sample is based on U.S. corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<15%	15%–25%	25%–35%	35%–45%	45%–55%	>55%	All
Intercept	0.009	0.008	0.018	0.014	0.041	0.065	0.022
	(3.461)	(3.237)	(5.972)	(3.611)	(5.652)	(7.849)	(12.284)
$\Delta Lev_{i,t}$	0.003	0.013	0.016	0.038	0.052	0.114	0.033
	(0.535)	(9.510)	(8.389)	(15.091)	(15.669)	(17.553)	(21.010)
ΔRF_t	-0.277	-0.386	-0.496	-0.719	-0.819	-1.094	-0.564
	(-22.319)	(-33.087)	(-28.819)	(-22.612)	(-13.504)	(-16.358)	(-43.590)
ΔRF_t^2	0.119	0.172	0.087	0.193	0.127	0.209	0.151
	(3.007)	(4.521)	(1.935)	(3.039)	(1.319)	(1.783)	(5.953)
$\Delta Slope_t$	0.269	0.364	0.447	0.633	0.825	0.870	0.504
	(13.547)	(23.687)	(17.824)	(13.341)	(9.216)	(9.455)	(28.223)
ΔVIX_t	0.003	0.005	0.008	0.009	0.009	0.008	0.006
	(1.393)	(7.146)	(5.326)	(5.404)	(2.862)	(2.529)	(8.511)
RM_t	-0.016	-0.019	-0.030	-0.041	-0.062	-0.089	-0.037
	(-10.064)	(-16.267)	(-20.129)	(-18.921)	(-16.231)	(-20.077)	(-38.037)
$\Delta Jump_t$	0.005	0.008	0.011	0.018	0.021	0.031	0.014
	(5.194)	(13.170)	(12.107)	(12.334)	(7.403)	(10.959)	(23.240)
Mean R^2	0.306	0.329	0.348	0.408	0.425	0.392	0.355
Median R^2	0.323	0.352	0.358	0.436	0.450	0.408	0.378
Obs.	81215	124240	97662	57243	32611	56817	449788
Bonds	1261	1900	1288	878	515	984	6826

Table III: **Principal Component Analysis**

For each industrial bond i with at least 25 monthly observations of yield spread changes, $\Delta YS_{i,t}$, I estimate the model:

$$\Delta YS_{i,t} = \alpha_i + \boldsymbol{\beta}_i^T \boldsymbol{\Delta S}_{i,t} + \varepsilon_{i,t},$$

where $\boldsymbol{\Delta S}_{i,t} := [\Delta Lev_{i,t}, \Delta RF_t, \Delta RF_t^2, \Delta Slope_t, \Delta VIX_t, RM_t, \Delta Jump_t]$ is the vector of the structural model variables, with $\Delta Lev_{i,t}$ as the change in firm leverage, ΔRF_t the change in 10-year Treasury interest rate, ΔRF_t^2 the squared change in the 10-year Treasury interest rate, $\Delta Slope_t$ the change in the slope of the term structure, ΔVIX_t the change in VIX_t index, RM_t the S&P 500 return, and $\Delta Jump_t$ the change in a jump factor based on S&P 500 index options. I then assign each month's residuals to one of 18 bins defined by three maturity groups (less than five years, five to eight years, and over eight years) and six leverage groups (less than 15%, 15%–25%, 25%–35%, 35%–45%, 45%–55%, and greater than 55%) and compute an average residual. I extract the principal components of the covariance matrix of these residuals. For each bin, I report the number of bonds, the number of observations, the principal components loadings, and the ratio of variation of the residual to the total variation. I further report the proportions of the variance of the residuals explained by the first and second principal components, PC1 and PC2, respectively, and the total unexplained variance of the regression in percentage points. The sample is based on U.S. corporate bond transaction data from TRACE for the period 2004–2021.

Leverage	Maturity	Bonds	Observations	PC1	PC2	Exp
1	1	913	36914	0.094	0.065	0.011
1	2	642	16399	0.096	0.114	0.011
1	3	702	27902	0.070	0.094	0.006
2	1	1331	51716	0.130	0.140	0.019
2	2	1052	25877	0.136	0.141	0.018
2	3	1151	46647	0.094	0.132	0.010
3	1	877	33402	0.185	0.084	0.035
3	2	785	20118	0.156	0.161	0.024
3	3	813	44142	0.129	0.124	0.017
4	1	622	21038	0.268	0.019	0.070
4	2	597	15212	0.235	0.261	0.053
4	3	539	20993	0.175	0.159	0.033
5	1	403	13519	0.329	-0.585	0.117
5	2	372	9503	0.309	0.162	0.090
5	3	262	9589	0.190	0.100	0.039
6	1	796	25848	0.440	-0.330	0.180
6	2	721	17301	0.401	-0.267	0.152
6	3	398	13668	0.326	0.465	0.115
Proportion of Variance				0.797	0.054	
Unexplained Variance					1.123	

Table IV: **Heterogeneity by Firm Fundamentals**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). I estimate the model with and without inflation-risk proxies separately for groups formed on firm fundamentals. Bonds are assigned to groups based on the average value of the relevant characteristic over the bond's sample life. Panel A reports results by firm size, Panel B by return on assets, Panel C by profit margin and Panel D by credit rating. Each panel reports average coefficients, cross-sectional t-statistics, mean and median adjusted R^2, and the number of observations and bonds. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

Panel A: Size

Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.432	0.466	102717	1647
Small	-0.363	0.605	0.072	0.460	0.508	102717	1647
	(-9.658)	(3.853)	(1.005)				
	-	-	-	0.512	0.537	128291	2016
2	-0.390	0.575	0.091	0.536	0.577	128291	2016
	(-11.629)	(6.218)	(1.328)				
	-	-	-	0.535	0.566	38890	568
3	-0.335	0.698	-0.094	0.560	0.603	38890	568
	(-6.487)	(4.262)	(-1.218)				
	-	-	-	0.536	0.553	102367	1507
4	-0.295	0.875	0.031	0.563	0.596	102367	1507
	(-13.125)	(9.698)	(1.021)				
	-	-	-	0.543	0.563	77523	1088
Large	-0.203	0.935	0.081	0.566	0.601	77523	1088
	(-7.105)	(8.180)	(1.591)				

Panel B: ROA

Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.517	0.544	90114	1468
Low	-0.819	1.147	0.033	0.538	0.578	90114	1468
	(-13.074)	(5.302)	(0.288)				
	-	-	-	0.551	0.593	89823	1336
2	-0.311	0.710	0.060	0.579	0.627	89823	1336
	(-14.070)	(8.400)	(1.834)				
	-	-	-	0.495	0.506	89987	1279
3	-0.208	0.615	0.077	0.526	0.556	89987	1279
	(-10.217)	(6.757)	(1.627)				
	-	-	-	0.486	0.522	89955	1379
4	-0.149	0.454	0.076	0.511	0.550	89955	1379
	(-10.220)	(6.391)	(1.481)				
	-	-	-	0.475	0.491	89909	1364
High	-0.111	0.616	0.037	0.499	0.533	89909	1364
	(-6.639)	(10.149)	(1.138)				

(Continued on the next page)

Panel C: Margin							
Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.521	0.556	90096	1446
Low	-0.818	1.086	-0.004	0.542	0.585	90096	1446
	(-13.197)	(5.061)	(-0.036)				
	-	-	-	0.515	0.542	89866	1286
2	-0.310	0.634	0.117	0.548	0.592	89866	1286
	(-12.082)	(6.051)	(2.690)				
	-	-	-	0.478	0.500	89903	1334
3	-0.212	0.533	0.064	0.506	0.559	89903	1334
	(-9.340)	(5.892)	(1.508)				
	-	-	-	0.506	0.532	89908	1346
4	-0.151	0.768	0.113	0.527	0.567	89908	1346
	(-8.752)	(10.973)	(2.086)				
	-	-	-	0.504	0.533	89926	1411
High	-0.123	0.547	0.000	0.528	0.562	89926	1411
	(-9.271)	(10.183)	(0.002)				

Panel D: Credit Rating							
Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.392	0.412	35615	546
AAA-AA	-0.052	0.474	-0.003	0.413	0.454	35615	546
	(-3.981)	(7.934)	(-0.073)				
	-	-	-	0.456	0.477	117438	1866
A	-0.102	0.452	0.070	0.481	0.512	117438	1866
	(-8.945)	(10.774)	(2.952)				
	-	-	-	0.531	0.565	171463	2847
BBB	-0.224	0.541	0.078	0.554	0.607	171463	2847
	(-15.226)	(8.845)	(3.575)				
	-	-	-	0.554	0.589	48983	976
BB	-0.580	0.990	0.053	0.578	0.628	48983	976
	(-9.939)	(5.296)	(0.730)				
	-	-	-	0.502	0.556	42649	892
B-C	-0.952	1.302	-0.007	0.524	0.594	42649	892
	(-9.331)	(3.714)	(-0.038)				

Table V: **Heterogeneity by Operating Channels**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). I estimate the model with and without inflation-risk proxies separately for groups formed on operating characteristics. Bonds are assigned to groups based on the average value of the relevant characteristic over the bond's sample life. Panel A reports results by operating leverage, Panel B by markup, and Panel C by input-cost exposure. Each panel reports average coefficients, cross-sectional t-statistics, mean and median adjusted R^2, and the number of observations and bonds. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

Panel A: Op. Leverage

Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.537	0.582	89998	1426
Low	-0.327	0.628	0.119	0.561	0.621	89998	1426
	(-10.597)	(4.343)	(1.542)				
	-	-	-	0.515	0.548	90002	1359
2	-0.380	0.858	0.029	0.546	0.596	90002	1359
	(-10.588)	(6.361)	(0.533)				
	-	-	-	0.483	0.496	89914	1312
3	-0.279	0.559	0.209	0.508	0.538	89914	1312
	(-9.797)	(6.113)	(2.956)				
	-	-	-	0.496	0.523	89958	1352
4	-0.288	0.912	-0.067	0.518	0.559	89958	1352
	(-8.670)	(7.293)	(-1.286)				
	-	-	-	0.492	0.509	89916	1377
High	-0.366	0.623	-0.007	0.516	0.549	89916	1377
	(-8.296)	(5.438)	(-0.095)				

Panel B: Markup

Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.526	0.570	89948	1428
Low	-0.481	0.785	0.014	0.552	0.608	89948	1428
	(-10.159)	(4.123)	(0.150)				
	-	-	-	0.505	0.538	89935	1354
2	-0.345	0.870	-0.002	0.536	0.588	89935	1354
	(-9.878)	(7.717)	(-0.025)				
	-	-	-	0.519	0.544	89938	1309
3	-0.326	0.805	0.128	0.545	0.579	89938	1309
	(-9.784)	(7.862)	(3.520)				
	-	-	-	0.485	0.507	90108	1371
4	-0.305	0.543	0.091	0.502	0.541	90108	1371
	(-9.952)	(6.532)	(1.495)				
	-	-	-	0.489	0.506	89770	1361
High	-0.178	0.591	0.054	0.516	0.548	89770	1361
	(-8.240)	(6.432)	(1.571)				

Panel C: Input-cost exposure

Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.520	0.575	33515	641
Low	-0.183	0.425	-0.050	0.544	0.600	33515	641
	(-4.626)	(2.522)	(-0.755)				
	-	-	-	0.481	0.530	34046	547
2	-0.183	0.352	0.158	0.513	0.558	34046	547
	(-6.406)	(2.187)	(0.994)				
	-	-	-	0.432	0.442	34359	455
3	-0.205	0.655	0.032	0.453	0.471	34359	455
	(-5.312)	(2.332)	(0.779)				
	-	-	-	0.467	0.484	34420	495
4	-0.317	0.561	0.126	0.489	0.496	34420	495
	(-3.615)	(2.659)	(1.605)				
	-	-	-	0.559	0.600	34259	557
High	-0.345	1.009	-0.033	0.580	0.642	34259	557
	(-8.336)	(6.130)	(-0.481)				

Table VI: **Heterogeneity by Debt Structure**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta\sigma^S_{i,t}$, $\Delta\rho^S_t$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). I estimate the model with and without inflation-risk proxies separately for groups formed on debt-structure characteristics. Bonds are assigned to groups based on the average value of the relevant characteristic over the bond's sample life. Panel A reports results by leverage, Panel B by refinancing intensity, Panel C by debt growth, Panel D separates firms with substantial floating-rate debt from firms that predominantly rely on fixed-rate debt, and Panel E reports results by maturity. Each panel reports average coefficients, cross-sectional t-statistics, mean and median adjusted R^2, and the number of observations and bonds. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

Panel A: Leverage

Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.457	0.483	81215	1261
<15%	-0.133	0.541	0.031	0.487	0.523	81215	1261
	(-6.832)	(6.091)	(0.769)				
	-	-	-	0.502	0.528	124240	1900
15%-25%	-0.136	0.570	0.087	0.527	0.563	124240	1900
	(-9.623)	(10.363)	(2.071)				
	-	-	-	0.494	0.503	97662	1288
25%-35%	-0.191	0.534	0.063	0.521	0.555	97662	1288
	(-9.186)	(5.418)	(1.772)				
	-	-	-	0.556	0.595	57243	878
35%-45%	-0.403	0.943	0.167	0.577	0.623	57243	878
	(-12.241)	(5.349)	(3.741)				
	-	-	-	0.556	0.581	32611	515
45%-55%	-0.618	0.579	0.051	0.587	0.616	32611	515
	(-10.176)	(2.346)	(0.445)				
	-	-	-	0.515	0.547	56817	984
>55%	-0.911	1.328	-0.077	0.534	0.588	56817	984
	(-10.682)	(5.166)	(-0.486)				

Panel B: Refinancing Intensity

Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.538	0.583	88982	1477
Low	-0.514	0.985	0.047	0.557	0.611	88982	1477
	(-10.302)	(6.705)	(0.435)				
	-	-	-	0.530	0.557	89020	1246
2	-0.297	0.627	0.019	0.550	0.588	89020	1246
	(-9.251)	(5.504)	(0.386)				
	-	-	-	0.517	0.545	89068	1278
3	-0.242	0.784	0.080	0.544	0.584	89068	1278
	(-9.096)	(5.629)	(1.968)				
	-	-	-	0.470	0.493	89069	1285
4	-0.268	0.658	0.099	0.504	0.538	89069	1285
	(-9.134)	(5.686)	(2.364)				
	-	-	-	0.470	0.489	88872	1476
High	-0.289	0.527	0.036	0.497	0.528	88872	1476
	(-10.952)	(5.544)	(0.720)				

(Continued on the next page)

Panel C: Debt Growth							
Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.506	0.541	90066	1610
Low	-0.260	0.784	0.080	0.533	0.589	90066	1610
	(-9.087)	(6.273)	(1.234)				
	-	-	-	0.505	0.523	89908	1195
2	-0.280	0.727	0.147	0.538	0.567	89908	1195
	(-11.779)	(8.506)	(2.179)				
	-	-	-	0.504	0.524	89919	1171
3	-0.276	0.605	0.133	0.530	0.559	89919	1171
	(-8.830)	(5.249)	(2.645)				
	-	-	-	0.508	0.537	89946	1296
4	-0.305	0.553	0.096	0.528	0.564	89946	1296
	(-10.388)	(6.406)	(2.544)				
	-	-	-	0.502	0.544	89949	1554
High	-0.495	0.856	-0.130	0.524	0.584	89949	1554
	(-10.236)	(5.323)	(-1.559)				

Panel D: Float							
Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.502	0.533	399704	6243
Fixed	-0.338	0.697	0.055	0.527	0.571	399704	6243
	(-20.093)	(11.533)	(2.040)				
	-	-	-	0.477	0.510	30662	672
Float	-0.148	0.497	0.053	0.492	0.523	30662	672
	(-4.787)	(3.587)	(0.701)				

Panel E: Maturity							
Group	$\Delta E[\mu^S]_{i,t}$	$\Delta\sigma^S_{i,t}$	$\Delta\rho^S_t$	Mean R^2	Median R^2	Obs.	Bonds
	-	-	-	0.442	0.454	165679	3698
<5	-0.238	0.357	0.066	0.473	0.498	165679	3698
	(-9.932)	(4.262)	(1.452)				
	-	-	-	0.518	0.562	75570	2251
5-8	-0.495	0.485	0.022	0.536	0.605	75570	2251
	(-11.957)	(3.542)	(0.436)				
	-	-	-	0.532	0.544	121874	1603
>8	-0.293	1.052	-0.019	0.549	0.581	121874	1603
	(-12.063)	(12.584)	(-0.841)				

Table VII: **Inflation Risk and Yield Spread Changes: All Bonds**

For each bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes inflation-risk proxies, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022) and Eisfeldt, Herskovic and Liu (2024). Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals, reporting the variance explained by the first two principal components and total unexplained variance. Panel C includes R^2 values, F-statistics, and p-values from a Wald-test of the time-series regression of PC1 on inflation-risk proxies. The sample is based on U.S. corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Individual Bond Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.458			-0.419	-0.311	-0.319	-0.252
		(-46.945)			(-45.521)	(-30.094)	(-29.269)	(-23.047)
$\Delta\sigma^S_{i,t}$			1.399		1.120	0.922	0.940	0.587
			(45.221)		(37.637)	(27.165)	(24.742)	(14.328)
$\Delta\rho^S_t$				0.378	0.098	0.028	0.011	0.021
				(22.919)	(6.589)	(1.476)	(0.542)	(0.958)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Mean R^2	0.339	0.370	0.401	0.374	0.431	0.501	0.513	0.525
Median R^2	0.358	0.390	0.406	0.385	0.439	0.531	0.549	0.569
Obs.	911634	911634	911634	911634	911634	911634	911634	911634
Bonds	14338	14338	14338	14338	14338	14338	14338	14338
Panel B: Principal Component Analysis								
FVE		0.146	0.197	0.098	0.336	0.260	0.272	0.257
PC1	0.765	0.737	0.732	0.763	0.716	0.723	0.717	0.683
PC2	0.135	0.143	0.138	0.130	0.142	0.112	0.116	0.131
UV	0.658	0.562	0.528	0.593	0.437	0.222	0.202	0.156
Panel C: Time-Series Regression of PC1 on Inflation-Risk Proxies								
Adj. R^2		0.083	0.134	0.044	0.160	0.088	0.078	0.040
R^2		0.088	0.138	0.048	0.172	0.101	0.092	0.054
F-stat		19.759	32.944	10.451	14.155	7.682	6.857	3.862
p-value		0.000	0.000	0.001	0.000	0.000	0.000	0.010
Obs.		208	208	208	208	208	208	208

Table VIII: **Inflation Risk and Yield Spread Changes: End of Month Only**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + \nu_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes inflation-risk proxies, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022) and Eisfeldt, Herskovic and Liu (2024). Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals, reporting the variance explained by the first two principal components and total unexplained variance. Panel C includes R^2 values, F-statistics, and p-values from a Wald-test of the time-series regression of PC1 on inflation-risk proxies. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Individual Bond Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.736			-0.619	-0.505	-0.517	-0.447
		(-15.113)			(-16.159)	(-9.109)	(-9.293)	(-8.719)
$\Delta\sigma^S_{i,t}$			2.016		1.519	1.190	1.105	0.362
			(13.332)		(12.026)	(8.417)	(7.563)	(2.559)
$\Delta\rho^S_t$				0.746	0.269	0.010	-0.066	-0.105
				(9.204)	(4.075)	(0.121)	(-0.779)	(-1.279)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Mean R^2	0.386	0.418	0.454	0.427	0.483	0.545	0.556	0.568
Median R^2	0.414	0.449	0.473	0.455	0.505	0.585	0.602	0.613
Obs.	236694	236694	236694	236694	236694	236694	236694	236694
Bonds	4356	4356	4356	4356	4356	4356	4356	4356
Panel B: Principal Component Analysis								
FVE		0.137	0.243	0.100	0.356	0.353	0.366	0.307
PC1	0.611	0.546	0.573	0.586	0.502	0.494	0.517	0.503
PC2	0.127	0.151	0.127	0.124	0.160	0.158	0.156	0.170
UV	4.005	3.457	3.032	3.605	2.581	1.485	1.232	1.134
Panel C: Time-Series Regression of PC1 on Inflation-Risk Proxies								
Adj. R^2		0.065	0.051	0.033	0.090	0.048	0.036	0.029
R^2		0.069	0.055	0.037	0.103	0.062	0.050	0.043
F-stat		15.354	12.088	8.017	7.785	4.463	3.569	3.047
p-value		0.000	0.001	0.005	0.000	0.005	0.015	0.030
Obs.		208	208	208	208	208	208	208